

The Compass



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Community Mathematics Centre

Schools of the Krishnamurti Foundation India
(KFI)

Statement of intent

The magazine is published by the Community Mathematics Centre. It is a collaborative initiative of the schools of the Krishnamurti Foundation India (KFI).

Its intention is:

- To provide material for mathematics and science teachers who teach at the high school or middle school levels, and for students who study at these levels.
- To serve as a platform where teachers and students can share their findings and discoveries with others who share their love for the subject.

In addition to mathematical content, in each issue we shall share a few ‘snippets’ from the world of science — short extracts abstracted from articles in online magazines, selected on the basis of intrinsic interest.

The Compass is published as an online magazine free of cost. Articles may be downloaded and used freely by teachers, but with proper attribution.

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Colour coding

The magazine has articles addressed to teachers who teach at different levels: Lower Primary level; Upper Primary and High School levels; and Senior Secondary and Junior College levels. There is also a Problem Corner. In addition we have the Snippets.

For the convenience of the reader, we use the following colour coding to distinguish between the various levels. Of course, such a division can never be exact. So, our apologies in advance for errors of judgement!

- Primary & Upper Primary (Classes 4, 5, 6, 7)
- Upper Primary & High School (Classes 6, 7, 8, 9, 10)
- Senior Secondary & Junior College (Classes 10, 11, 12)
- Problem Corner
- Snippets

Playing with Primitive Pythagorean Triples – 4

We continue our exploration of Primitive Pythagorean Triples (PPTs). In this article we describe three connections:

- (a) Between PPTs and incircles of triangles;
- (b) Between PPTs and complex numbers — a connection that at first sight seems quite surprising;
- (c) Between PPTs and “Fermat’s last theorem.”

The first of these ‘connections’ leads us into an interesting side-exploration that has nothing to do with PPTs. We will describe this later in the article.

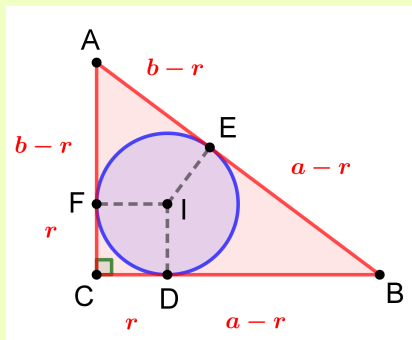
Incircles

We prove the following simple and appealing result.

Theorem. *The in-radius of a right-angled triangle with integer sides is an integer.*

(The in-radius of a triangle is simply the radius of its incircle.)

Proof of the theorem



Let $\triangle ABC$ be right-angled at C as shown. Let its sides be a, b, c ; so (a, b, c) is a PPT.

Let its incircle, centre I , touch the sides at D, E, F . Let the radius of the incircle be r . Observe that $CDIF$ is a square with side r .

Figure 1: A right-angled triangle with its incircle

We have, $BD = a - r$ and $AF = b - r$. Making use of the property that the two tangents to a circle from a point outside the circle have equal length, we see that $BE = a - r$ and $AE = b - r$. Since $AB = AE + BE$, we have $c = (b - r) + (a - r)$, giving

$$r = \frac{a + b - c}{2}.$$

We had proved earlier that if (a, b, c) is a PPT, then a and b have opposite parity (hence $a + b$ is odd), and c is odd. This implies that $a + b - c$ is even, hence r is an integer. ■

The r -values for a few PPTs are displayed in the table below.

PPT	(3,4,5)	(5,12,13)	(8,15,17)	(7,24,25)	(9,40,41)	(12,35,37)
r	1	2	3	3	4	5

Problems for exploration

1. Find all PPTs (a, b, c) for which $r = 1$.
2. Find all PPTs (a, b, c) for which $r = 2$.
3. Find all PPTs (a, b, c) for which $r = 3$.
4. Find all PPTs (a, b, c) for which $r = 4$.
5. Find a general procedure for listing all PPTs that correspond to a given value of r .

Sample analysis

Just for fun, let us fully analyze the first problem: to find all PPTs (a, b, c) for which $r = 1$.

We have already shown that $r = \frac{1}{2}(a + b - c)$. Hence if $r = 1$, it must be that $a + b - c = 2$, i.e., $c = a + b - 2$.

So the Pythagorean equation becomes:

$$a^2 + b^2 = (a + b - 2)^2.$$

Expanding and bringing all the terms to one side we get

$$2ab - 4a - 4b + 4 = 0.$$

Hence,

$$ab - 2a - 2b + 2 = 0.$$

We must find all pairs (a, b) of positive integers satisfying the above equa-

tion. We shall do so using a familiar trick: factorization. We write,

$$ab - 2a - 2b + 2 = 0,$$

$$\therefore ab - 2a - 2b = -2,$$

$$\therefore ab - 2a - 2b + 4 = -2 + 4 = 2,$$

$$\therefore (a - 2)(b - 2) = 2.$$

We must solve the equation $(a - 2)(b - 2) = 2$ for a, b , over the positive integers $\mathbb{N}$. We clearly require $a > 2$ and $b > 2$.

Now 2 can be expressed as a product of two positive integers in just one way: $2 = 1 \times 2$. Hence there are just two possibilities for the solution:

$$a - 2 = 2 \text{ and } b - 2 = 1, \text{ giving } (a, b) = (4, 3),$$

or,

$$a - 2 = 1 \text{ and } b - 2 = 2, \text{ giving } (a, b) = (3, 4).$$

Either way we get $c = 5$, and this corresponds to the PPT $(3, 4, 5)$.

So, $(3, 4, 5)$ is the only PPT for which the corresponding in-radius is 1.

You may want to explore this question:

Is there any other value for the in-radius r corresponding to
which there is just one PPT?

Remark

Here is another way of proving the formula $r = \frac{1}{2}(a + b - c)$.

We know that the area K and the semi-perimeter s of a triangle are related to the radius r of the incircle as follows:

$$K = rs.$$

For the right-angled triangle corresponding to the PPT (a, b, c) , we have,

$$K = \frac{1}{2}ab, \quad s = \frac{1}{2}(a + b + c).$$

Hence:

$$\begin{aligned} r &= \frac{K}{s} = \frac{ab}{a + b + c} \\ &= \frac{ab}{a + b + c} \cdot \frac{a + b - c}{a + b - c}. \end{aligned}$$

Now,

$$\begin{aligned} (a + b + c)(a + b - c) &= (a + b)^2 - c^2 \\ &= a^2 + 2ab + b^2 - c^2 \\ &= 2ab, \end{aligned}$$

since $a^2 + b^2 = c^2$. Therefore,

$$r = \frac{ab(a + b - c)}{2ab} = \frac{a + b - c}{2},$$

as earlier.

PPTs and complex numbers — a strange connection

Recall that complex numbers are entities $a + bi$ where i is defined by the relation $i^2 = -1$ and a, b are real numbers; a is called the *real part* and b

the *imaginary part* of $a + bi$. The set of all possible complex numbers is generally denoted by $\mathbb{C}$. We assume that readers are familiar with the algebra of complex numbers; i.e., how to add, subtract, multiply, and divide by complex numbers.

Our interest will only be in complex numbers whose real and imaginary parts are both integers. Such entities are called *Gaussian integers*. They are of great interest to number theorists. We will study them in greater detail later.

Here is the connection between PPTs and complex numbers. Select any Gaussian integer $z = a + bi$ with $a \neq b$; compute its square, z^2 . Also compute the norm (or magnitude) of z^2 . From the answers, we directly obtain a Pythagorean triple. (If there are negative numbers in z^2 , drop the sign.) If we also ensure that a, b are coprime and have opposite parity, then we obtain a PPT. And that's it!

Examples

- Let $z = 1 + 2i$. Square it and compute the norm of the answer:

$$(1 + 2i)^2 = 1 + 4i + 4i^2 = -3 + 4i,$$

$$|-3 + 4i| = \sqrt{(-3)^2 + 4^2} = 5.$$

This yields the familiar PPT (3,4,5).

- Let $z = 2 + 3i$. Square it and compute the norm of the answer:

$$(2 + 3i)^2 = 4 + 12i + 9i^2 = -5 + 12i,$$

$$|-5 + 12i| = \sqrt{(-5)^2 + 12^2} = 13.$$

This yields another familiar PPT (5,12,13).

- Let $z = 1 + 4i$. This yields:

$$(1 + 4i)^2 = 1 + 8i + 16i^2 = -15 + 8i,$$

$$|-15 + 8i| = \sqrt{(-15)^2 + 8^2} = 17.$$

This yields the PPT (15,8,17).

- Let $z = 8 + 3i$. This yields:

$$(8 + 3i)^2 = 64 + 48i + 9i^2 = 55 + 48i,$$

$$|55 + 48i| = \sqrt{55^2 + 48^2} = 73.$$

This yields the PPT (55,48,73).

Questions for exploration

1. Why does this recipe work?
2. Why do we require the condition that the real and imaginary parts of z must be coprime and of opposite parity? What happens if either of these conditions does not hold?

Pythagorean triples and Fermat's last theorem (FLT)

To close, we mention an amusing connection between the topic of PTs and a particular case of Fermat's last theorem ('FLT' as it is generally known).

The reader will surely know the statement of FLT:

For any integer n greater than 2, there do not exist positive integers x, y, z such that $x^n + y^n = z^n$.

The statement of FLT is compact and very easy to grasp, but the theorem is amazingly difficult to prove. The simplest case of the theorem is $n = 4$, but even this presents a major challenge! Fermat himself found a proof for this case ($n = 4$) using a novel approach that he had pioneered, now called the “method of infinite descent”.

We see from FLT that there do not exist any PPTs (a, b, c) such that all three of a, b, c are perfect squares. For, if there were such a PPT, it would yield a solution in positive integers to the equation $x^4 + y^4 = z^4$.

But a stronger statement can be made: there do not exist any PPTs (a, b, c) such that both a and b are perfect squares. This conclusion drops out from Fermat’s proof of FLT for the case $n = 4$; we obtain it as a bonus result.

Problems for exploration

We had mentioned earlier that in studying the question about the radius of the incircle of a right-angled triangle with integer sides, related questions emerge naturally. For example,

Which integer-sided triangles have incircles with integer radius?

And:

Find all integer-sided triangles whose incircle has radius $r = 1$
(or any other given value).

We leave this exploration for the reader to pursue.

The Big Wide World of Data

Visualisation – 5

As stated in earlier issues, this column aims to present a running series of case studies and/or short articles on the theme of data visualisation. We shall present a few striking examples of this in each issue (with little or no commentary).

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From the article “Gen Z are less likely to favour fully remote working compared with millennials and boomers” by Millie Giles, available at https://sherwood.news/culture/gen-z-less-likely-favor-full-remote-working-compared-millennials-boomers/?utm_source=chartr&utm_medium=email&utm_campaign=chartr_20250806

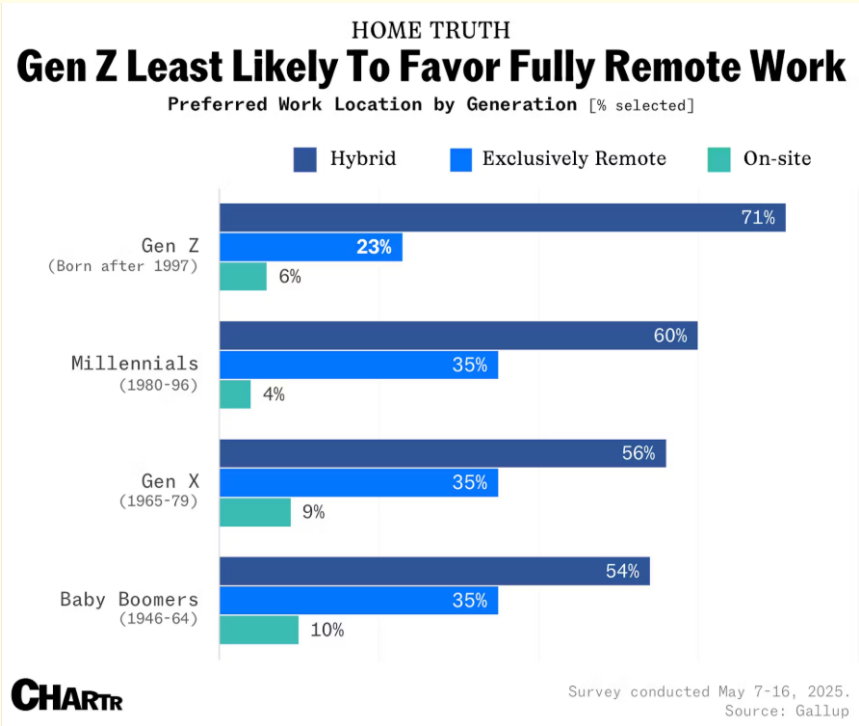


Figure 2: Source: <https://tinyurl.com/mr2b7rxx>

Per a new Gallup survey, it turns out young workers do want to go back to the office ... sometimes.

Since the pandemic, working from home has not only endured as a pretty standard practice for office-based companies, but also remains a requisite for some job seekers accustomed to having at least some days where their commute time is measured in seconds rather than minutes or hours.

Now, several years on from a string of Covid-induced lockdowns, the novelty of doing one's job in a room of one's own may have finally worn off ... especially for the youngest cohort of employees.

New kids on the (office) block

A recent Gallup survey found that fully remote work was least popular with Gen Z among all age groups surveyed, with only 23% saying they preferred working from home full time — considerably less than the 35% of millennial, Gen X, and baby boomer respondents who favoured the wholly WFH model.

Gallup outlined one factor that could be driving a desire to return to the office: Gen Z is statistically the loneliest generation. The same survey found that 27% of Gen Z reported feeling lonely the day before — almost 3x the share of boomers.

Even so, while a large faction of Gen Z don't want to be at home permanently, they definitely don't want to be in the office all the time either, with only 6% of Gen Z preferring to be totally on-site. Indeed, most seem to want options: while hybrid was the preferred work setup across all generations, it was by far the top pick for younger participants (71%).

Another reason for Gen Z's office push could be that being in a co-working environment might translate to better engagement. An earlier Gallup poll, cited by Business Insider in January, found that only 30% of workers under 35 years old reported feeling engaged at work last year — dipping below the older cohort for the first time in the survey's history.

Jayant Narlikar, 1938–2025

Jayant Vishnu Narlikar, a pioneering Indian astrophysicist, passed away at the age of 86, leaving behind a legacy that reshaped the understanding of the universe and popularized science in India. Known for his contributions to cosmology, Narlikar co-developed the Quasi-Steady State Cosmology Theory with Fred Hoyle, which challenged the dominant Big Bang Theory. While the Big Bang Theory posits that the universe began with a massive explosion and has been expanding since, Narlikar's theory suggested that the universe has always existed in a constant state of density, without a singular origin.

Born on July 19, 1938, in Varanasi, Narlikar was deeply influenced by his father, a professor at Banaras Hindu University. He pursued his BSc at BHU and later studied mathematics at Cambridge, where he began collaborating with Hoyle.

Narlikar's contributions extended beyond theoretical physics; he was instrumental in advancing astrophysical research in India. After returning to India in 1972, he worked at the Tata Institute of Fundamental Research and founded the Inter-University Centre for Astronomy and Astrophysics

(IUCAA), the first government-supported institute in the country solely dedicated to astrophysics.



Figure 3: Jayant Narlikar. Photograph by Biswarup Ganguly, CC BY 3.0, <https://commons.wikimedia.org/w/index.php?curid=14985638>

Narlikar was not just a scientist but also a passionate science communicator. He authored around 10 books in Marathi, making complex topics like space science and cosmology accessible to the masses, including children. His efforts in science education were significant; he served as Chairman for the NCERT advisory committee for science and mathematics textbooks in 2005. His book *Akashashi Jadale Nate* ("Developing a Relationship with the Sky") epitomized his mission of bringing science closer to the public.

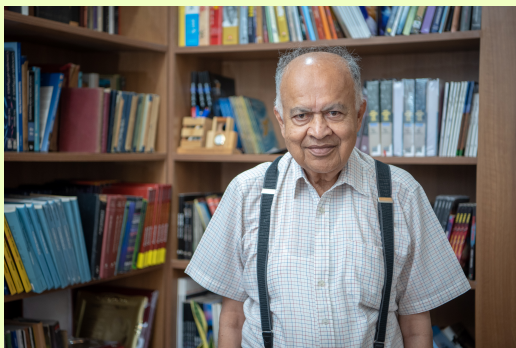


Figure 4: Jayant V. Narlikar in his IUCAA office. Source of photograph: <https://www.iucaa.in/en?Itemid=188>

Narlikar's influence extends to generations of Indian scientists, particularly in Maharashtra. Prominent astrophysicists, including his students and colleagues, credit him with inspiring their careers. He was known for providing his students with intellectual freedom, a quality that established IUCAA as a leading hub for astrophysics research. Despite his towering presence in the field, Narlikar was approachable and committed to fostering curiosity and inquiry.

Although the Big Bang Theory gained acceptance due to evidence like cosmic microwave background radiation, Narlikar's work remains significant for its role in stimulating cosmological research and encouraging alternative views about the universe's origins. His dual legacy as a scientist and educator has left an indelible mark on Indian astrophysics and science communication. As Dr. Sule aptly noted, Narlikar helped India "develop a relationship with the sky," inspiring a generation to explore the cosmos. (Essay based on [1].)

Appendix 1: The Steady State Cosmology Theory

The Steady State Cosmology Theory, proposed by Fred Hoyle and further developed by Jayant Narlikar, posits that the universe has no beginning or end and exists in a constant, unchanging state. Unlike the Big Bang Theory, which suggests the universe originated from a singular explosion and has been expanding since, the Steady State Theory argues that the universe is eternal and maintains a constant density despite its expansion. This is explained by the continuous creation of matter to compensate for the expanding space, ensuring a steady appearance over time.

In 1993, Hoyle and Narlikar extended this idea into the Quasi-Steady State Cosmology Theory, which introduced periodic episodes of matter creation, allowing for variations in density. Although the Big Bang Theory became widely accepted due to evidence like cosmic microwave background radiation, the Steady State Theory remains significant for promoting alternative cosmological inquiries and challenging conventional ideas.

Appendix 2: The Big Bang Cosmology Theory

The Big Bang Theory was first proposed by Belgian physicist and Catholic priest Georges Lemaître in 1927. He referred to it as the *hypothesis of the primeval atom* or the *cosmic egg*, suggesting that the universe began from a single, dense point that expanded to form the cosmos we see today.

Lemaître's ideas were supported and developed further by observational evidence, such as Edwin Hubble's discovery in 1929 that galaxies are moving away from each other, indicating an expanding universe. The term 'Big Bang' itself was coined in 1949 by Hoyle, ironically as a dismissive remark during a BBC radio broadcast.

The Big Bang Theory is the prevailing scientific explanation for the origin and evolution of the universe. It proposes that the universe began approximately 13.8 billion years ago as a singularity—an extremely hot, dense point—and has been expanding ever since. This expansion created space, time, matter, and energy.

The theory is supported by key observational evidence, including the detection of cosmic microwave background (CMB) radiation, which is the afterglow of the initial explosion, and the observed redshift of galaxies, indicating that the universe is still expanding. It also explains the abundance of light elements like hydrogen and helium, formed during the first few minutes after the Big Bang.

The Big Bang Theory does not address what might have caused the singularity or what preceded it. Nor does it say anything definitive about what happened in the immediate aftermath of the Big Bang. But it provides a robust framework for understanding the universe's large-scale structure, its evolution, and its ongoing expansion.

Further reading

- [1] Akanksha Mishra, "Indian astrophysics giant Jayant Vishnu Narlikar reshaped our relationship with the sky" from *The Print*, 20th May 2025, <https://theprint.in/science/indian-astrophysics-giant-jayant-vishnu-narlikar-reshaped-our-relationship-with-the-sky/2631621/>

Mangala Narlikar changed how girls in India learn math. She took the fear out of it

Adapted from: “Mangala Narlikar changed how girls in India learn math”, <https://tinyurl.com/4ycudwdn>. Author: Sakshi Mehra (edited by Ratan Priya)

In a generation when women's aspirations were often softened to fit domestic frames, Mangala Narlikar refused to choose between academic pursuit and care-giving.

Mangala Narlikar, a distinguished mathematician, led a life of quiet yet profound impact, balancing academic rigour with domestic responsibilities. Her career spanned institutions like the Tata Institute of Fundamental Research (TIFR), Cambridge, and Pune University, while her personal life revolved around raising three daughters and supporting her astrophysi-

cist husband, Jayant Narlikar. Despite societal expectations that often forced women to choose between family and career, Narlikar seamlessly integrated both, refusing to compromise on her intellectual pursuits.

Born into an era when women's aspirations were often curtailed, Narlikar excelled in mathematics, earning her PhD in analytic number theory in 1981 while raising a family. Her academic journey, though marked by part-time roles, was underscored by determination and a desire to prove herself in a male-dominated field. She consistently excelled in her studies and rejected the stereotype that women were unsuited for mathematics.



Figure 5: Mangala Narlikar. Source: By Arya Joshi - Own work, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=63323250>

Marriage to Jayant Narlikar brought challenges and opportunities. While her research was interrupted during their time in Cambridge, she resumed

her academic work upon returning to Mumbai in the 1970s. Living near TIFR allowed her to contribute part-time to research and teaching. Over the decades, she held guest lectureships, authored numerous books in English and Marathi, and influenced school curricula in Maharashtra. Her work with the Balbharati textbook board focused on making mathematics accessible and engaging, particularly for underprivileged students. As Chairperson of Balbharati in 2013, she introduced child-friendly features like visual aids and interactive problems, ensuring affordability and inclusivity.

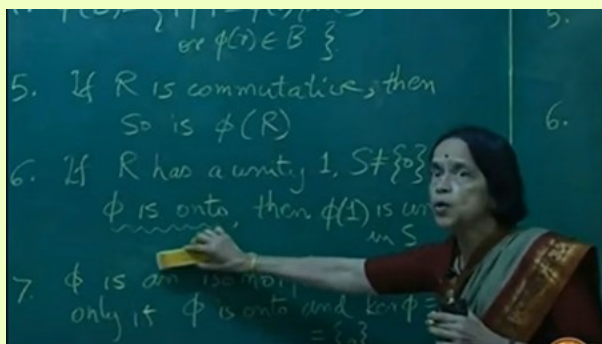


Figure 6: Mangala Narlikar lecturing

Narlikar's commitment to education extended beyond academia. She tutored the children of her domestic worker and later taught girls in slum communities, fostering a love for mathematics in those often marginalized. Her books, such as *An Easy Access to Basic Mathematics* and the award-winning Marathi title *Gargi Ajun Jeevant Aahe*, reflected her clarity and empathy as an educator.

Her personal and professional lives were deeply intertwined, with her home serving as a foundation for her legacy. All three of her daughters pursued careers in science and technology, a testament to her influence as both a mother and a mathematician. Her partnership with her husband was characterized by mutual respect and shared values, with Mangala's quiet strength supporting Jayant's work even as she pursued her own goals.

Mangala Narlikar passed away on July 17, 2023, after a prolonged battle with cancer. Her life, described as "part-time" by herself, was anything but incomplete. With her contributions to mathematics, education, and societal change, she left an enduring legacy that touched countless lives, demonstrating that intellect and compassion can coexist harmoniously.

(This is a summary of the original article, produced using GPT-4o.)

Eureka!

Article contributed by Anushka Rao

For the past few months, I have been running maths circles for two groups of 8-and-9-year-olds in an Oxford primary comprehensive school. The idea of the circles is reminiscent of the Russian circles for high school children that are popular in several parts of the world, where a small group of high achieving students get together with an ‘expert’ — a University professor or maths teacher — to work on problems outside the school curriculum. They work in ways that are different from usual school maths lessons and more like the experience of maths at university. There is a conversational and exploratory element to the circles, where the children work together, suggesting ideas and making conjectures, steered gently by the ‘expert’. There was an idea to start a similar movement with primary children here in the UK and a handful of primary maths circles exist across the country, two of which I run in Oxford.

Here is an account of a game we have been playing in the circles called *Eureka*. The game is explained by John Mason [1], Honorary Research Fellow and Professor Emeritus at Oxford University. Mason has spent over

40 years at the Open University, supporting the teaching of mathematics. He authored a wide range of papers, and dozens of Open University Books and teaching materials. I had read about this game and played it with some of my classes in Rishi Valley when I was a teacher there a few years ago, and I asked him for a reminder of how it works. This is what he said.

The Eureka Game was described by Martin Gardner in Scientific American sometime in the 1960s.

The version I recall goes like this.

I specify a triple of numbers and write down a property which the triple satisfies. . I then invite people to propose other triples, and I indicate with emojis whether the proposed triples satisfy the property. The Only Rule is that you never announce what the property is. If someone thinks they know it, then they announce EUREKA and contribute examples that they think will help others reach a conclusion. Often people discover that they have not quite determined the property when they offer examples they think will succeed, and ones that will fail.

It mirrors scientific enquiry because nature never tells us when we are 'right'.

Classic example: I offer 2,4,8 as my starting triple, but the property is "strictly increasing". It often takes people a while to let go of the implied multiplicative relationships!

It is close to the Silent Way, a teaching approach by Caleb Gattengo [2]. There the teacher offers one or more examples that satisfy some rule or property, in silence, and invites others to come up to the board and offer examples. The teacher uses a smiley or sad emoji to indicate

whether the example does or does not satisfy the rule/property. Again, when someone thinks they know the rule/property they offer examples which they think will help others to determine the rule. Some people do announce the rule eventually, others don't.

Let me know how it goes! I would use it briefly in a series of lessons so that experience builds up. Great for working on questioning your assumptions and making conjectures to test your hypotheses.

I was unable to find the article by Garner that Mason referred to. However, when I mentioned it to Shailesh Shirali, he recalled another game by Gardner called Eleusis that has similar principles. This was described in Gardner's column Mathematical Games, published in Scientific America, October 1977 [3]. Both games require inductive and deductive reasoning: the aim is for players to discover a secret rule by both inductively using information given by players and forming hypotheses about the rule and then deductively testing these ideas in subsequent play.

After a few weeks of playing the game, I wrote this account back to Mason:

My maths circles and I have been having such fun with Eureka.

I decided to start with the strictly increasing triple, which had everyone in a tizzy.

After an explanation of the game, I began with the triple, 1,3,7 which I wrote on the board. Almost at once, a child, Toby (names changed), said eureka! I tried to dissuade him, saying he couldn't possibly have reached certainty with one example, but he was adamant, especially when his example triple (2,4,8) proved valid. He is quite stubborn and dare I say often hubristic, so I decided against pushing back and asked instead for more examples.

The others, to my amusement, followed suit, excitedly saying eureka one after another when their examples received smiley faces.

(Examples included 4,8,16 and 5,7,9). Their folly was they were not paying much attention to the examples of others, just their own. One of them finally did and was confused, saying “that shouldn’t be a smiley face” (it took a while for them to remember not to talk!) but when I confirmed it, they started falling like dominoes, withdrawing their eureka.

The realisation that the data from their fellow explorers was useful was wonderfully evident to them. It was a good triple to begin with for this and so many other reasons. For instance, they were also cautious now, the idea of testing their conjectures, to get several smiley faces, before saying the word. They now recognised eureka as a precious thing. Another related reason why I think this was a good one to start with, was that they had an appreciation to not quickly jump to the conclusion that something would always work — which I think is a lovely introduction to a feeling of proof. They had initially jumped to the first relationship they saw to ‘fit’ the rule, but now realised they needed more before their moment of eureka.

It went on for a good 15 minutes, but they were captivated throughout; although they were struggling to find the rule, what kept their spirits up was elation at having a new idea, then testing it with anticipation. Because their enthusiasm didn’t wane, it gave me the confidence to hold on, although I was despairing inside!

Finally one of them, Sally, seemed to have got it. She tested several before saying eureka. I reminded her to give examples now that could help the others, and she rather brilliantly, first gave an example

quickly followed by a non-example (3,4,5 and 5,4,3). And after some thought said 2, 500, 1 million! She then gave more like these, where each number in the triple was wildly bigger than the previous. This broke the dam, and the eureka's flowed! Only after checking by using examples and non-examples, which again I thought was fab. Poor Jo was the only one without eureka, and his eyes began filling up so I whispered a hint to him.

It was such a hit that they have asked for it at every session and we now have a ritual where we end with a game. There was resistance and annoyance from the children when I insisted no one spoke the rule, that it was enough to test it and feel secure in our eureka's. But now there's a feeling of pride, like it's a secret code, one that doesn't need to be spoken but that all in the group can be confident each other knows.

What is always heartening is the carefully thought through examples of the have-eureka's to help out the others. For instance, for the triple — a number, its double add 1, its double add 1 — Wazeed (after reaching eureka) came up with 100,201,403, which he asked to change to $402 + 1$. The structure seemed to shine out to the children. [There is also often a surprising triple, just like one might find in a scientific enquiry. When I demonstrated this game with this rule to a group of maths educationalists in a working group I am part of, one of them offered the triple $-1, -1, -1$ which delightfully follows the rule!]

There are tears sometimes, particularly at the beginning when there was much gloating by the have-eureka's. But 8-year-olds can be amazingly empathetic as well, and a little speech from me about

trying to imagine what it felt like when everyone was shouting eureka and you didn't know what was going on, led to the children coming up with ideas of how to make it less stressful.

Mason replied to my account and observed:

I suspect that these experiences might be having more significance for the children than loads of ordinary arithmetic tasks! Really appreciating what it is like to do mathematics (and science) could have even wider ranging effects when they meet people with different perspectives on political, climate and other matters.

Here are a few other triples I have used in the circles, not in order of when I used them:

strictly increasing/decreasing, double the previous and add 1 (and variants to this, $3n - 1$, $5n + 2$), the Fibonacci sequence, halving/trebling etc the previous number, three numbers that add to 5, three numbers that multiply to 24, one more than a square number, one more than a prime, two odds and an even.

Further reading

[1] About Prof. John Mason:

<https://www.education.ox.ac.uk/person/john-mason/>

[2] Wikipedia, "Silent Way" from

https://en.wikipedia.org/wiki/Silent_Way

[3] Scientific American, 'The New Eleusis' (October 1977, Volume 237, Issue 4) from <https://www.scientificamerican.com/artic>

[le/mathematical-games-1977-10/](https://www.math.tamu.edu/outreach/mam/Elleusis-Gardner.pdf). For the complete article:
<https://www.math.tamu.edu/outreach/mam/Elleusis-Gardner.pdf>

About the author

Anushka Rao grew up in south India and studied in Rishi Valley School, KFI. She moved to the UK after completing an MSc (Mathematics). She trained to be a teacher at Oxford University in 2011 and completed an MA in Maths Education. She taught in schools in and around Oxford as well as at Rishi Valley School. During Covid and for a year after, she worked with an online startup, creating courses for teachers to teach maths with well-being.

She now teaches as a Lead Practitioner of Maths at Oxford Spires Academy where her job consists of teaching across age groups and mentoring new teachers. She also runs weekly maths circles for primary children (8- and 9-year-olds) with Primary Maths Circles. This is a fairly new initiative in UK schools, based on the model of maths circles for older children that has been around in Eastern Europe, Russia, and the USA. The 'circles' consist of small groups of children who work collaboratively to solve 'big' problems largely outside of the school curriculum.

She lives in Oxford with her husband who translates Buddhist scriptures from Tibetan, and three children. She may be contacted at anushkarao@gmail.com.

A Cube Difference Theorem: Rediscovery of Third-Order Finite Differences

Article contributed by Chetan Jha

Cubes of natural numbers — i.e., the numbers $1^3 = 1$, $2^3 = 8$, $3^3 = 27$, and so on — have many interesting features. Out of curiosity, I examined the gaps (differences) between successive cube numbers to see if any consistent pattern emerged. This led me to a surprising result: [the third-order difference between consecutive cube numbers is always 6](#).

I later found that this is a known result, falling under **Newton's finite difference theory**. But I discovered this on my own. So it was an independent rediscovery.

The cube number sequence

Here is a list of cube numbers from 1^3 to 12^3 .

n	1	2	3	4	5	6	7	8	9	10	11	12
n^3	1	8	27	64	125	216	343	512	729	1000	1331	1728

The sequence of first differences

The numbers $7 = 8 - 1$, $19 = 27 - 8$, $37 = 64 - 27$, $61 = 125 - 64$, ... which are the differences between successive cubes are the **first differences** of the cube sequence. We use the notation $\Delta^1(n^3)$ for these differences.

Here is a table of the first differences of the cube numbers.

n	1	2	3	4	5	6	7	8	9	10	11	12
$\Delta^1 n^3$	7	19	37	61	91	127	169	217	271	331	397	469

The sequence of second differences

The numbers $12 = 19 - 7$, $18 = 37 - 19$, $24 = 61 - 37$, $30 = 91 - 61$, ... which are the differences between the successive first differences are the **second differences** of the cube sequence. We use the notation $\Delta^2(n^3)$ for these differences.

The sequence of third differences

Similarly, the numbers $6 = 18 - 12$, $6 = 24 - 18$, $6 = 30 - 24$, $6 = 36 - 30$, ... which are the differences between the successive second differences are the **third differences** of the cube sequence. We use the notation $\Delta^3(n^3)$ for these differences.

The difference sequences for the cube numbers

The array below displays the cube numbers along with their first, second, and third differences:

n	1	2	3	4	5	6	7	8	9	10	11	12
n^3	1	8	27	64	125	216	343	512	729	1000	1331	1728
$\Delta^1(n^3)$	7	19	37	61	91	127	169	217	271	331	397	469
$\Delta^2(n^3)$	12	18	24	30	36	42	48	54	60	66	72	78
$\Delta^3(n^3)$	6	6	6	6	6	6	6	6	6	6	6	6

We observe that the third differences are all the same (equal to 6).

Theorem (Cube Difference Theorem). *For the sequence of consecutive cube numbers, the third-order finite difference is always equal to 6. That is,*

$$\Delta^3(n^3) = 6 \quad \text{for all } n.$$

Proof of the theorem

We have,

$$(n+1)^3 - n^3 = 3n^2 + 3n + 1.$$

This shows that the first differences of the cube numbers are described by the quadratic function $3n^2 + 3n + 1$. So,

$$\Delta^1(n^3) = 3n^2 + 3n + 1.$$

Taking the next difference,

$$\begin{aligned} \Delta^2(n^3) &= (3(n+1)^2 + 3(n+1) + 1) - (3n^2 + 3n + 1) \\ &= 6n + 6. \end{aligned}$$

So the second differences of the cube numbers are described by the linear function $6n + 6$. Note that these are consecutive multiples of 6, starting with 12.

Taking the next difference,

$$\Delta^3(n^3) = 6.$$

Hence, the third-order difference of n^3 is a constant, 6. ■

Historical Context

This result is of relevance in the theory of finite differences of polynomials. It occurs in studying the following:

- Newton's Forward Difference Formula
- Polynomial interpolation
- Discrete calculus

The general principle states the following:

Theorem. *For any polynomial of degree n , the n -th finite difference is a constant.*

In our case we are dealing with a cubic polynomial ($n = 3$), so we expect the third order difference to be a constant. We have found that to be the case (the constant being 6).

Closing comment

I feel happy that I discovered this mathematical pattern by myself and was also able to prove it.

I feel it is important for students like me to have such opportunities for free exploration. It encourages us to engage with mathematics beyond the syllabus.

I thank my teachers, parents for encouraging my curiosity and helping me organize my ideas in a formal structure. I dedicate this work to all students who observe patterns and set out boldly to question, test, and prove.

About the author

Chetan Jha is a student currently studying in Class 9 in India. He describes himself as a passionate mathematics enthusiast who spends his time exploring concepts beyond the class and discovering surprising mathematical relationships in different fields, from algebra and geometry to patterns and logic. He enjoys solving challenging problems. He may be contacted at arch7383@gmail.com

Miscellaneous Problems

About Primes

In this article we pose and solve a few problems about prime numbers. Please try your hand at the problems before reading the solutions.

Problem 1

The product of three prime numbers equals 43 times their sum. What are the three prime numbers?

Problem 2

Find all prime numbers p such that $p^2 + 2$ is prime.

Problem 3

Find all positive integers n such that $n^4 + 4$ is prime.

Problem 4

Find the largest prime divisor p of the binomial coefficient $\binom{2000}{1000}$, which is the central number in the 2000-th row of the Pascal Triangle.

Problem 5

- (a) Show that the integer $n^5 + n^4 + 1$ is not prime for any integer $n > 1$.
(b) Show that the integer $n^{101} + n^{37} + 1$ is not prime for any integer $n > 1$.

Problem 6

Find all natural numbers n with the following property: “Every natural number less than n can be expressed in exactly one way as the sum of distinct divisors of n .”

Please try out the problems
before reading on!

Solution to Problem 1

Let the prime numbers be p, q, r . The stated condition then translates to

$$pqr = 43(p + q + r).$$

As 43 is a divisor of the quantity on the right side, it must divide pqr too. And since $p, q, r, 43$ are all primes, one of p, q, r must be equal to 43. Let us suppose that $r = 43$. The resulting equation is then

$$pq = p + q + 43.$$

We must find prime numbers p, q that satisfy this equation. Without any loss of generality we may assume that $p \leq q$. The above equation may be rewritten as

$$pq - p - q = 43,$$

$$\therefore pq - p - q + 1 = 44,$$

$$\therefore (p - 1)(q - 1) = 44.$$

What are the different ways in which we can write 44 as a product of two positive integers? They are: $1 \times 44, 2 \times 22, 4 \times 11$. Hence,

$$(p - 1, q - 1) = (1, 44) \quad \text{OR} \quad (2, 22) \quad \text{OR} \quad (4, 11),$$

giving

$$(p, q) = (2, 45) \quad \text{OR} \quad (3, 23) \quad \text{OR} \quad (5, 12).$$

Since both p, q are prime, it must be that $p = 3, q = 23$. So we have our answer: the prime numbers are 3, 23 and 43. Let's check that these values satisfy the stated condition:

$$3 \times 23 \times 43 = 69 \times 43, \quad 43(3 + 23 + 43) = 43 \times 69.$$

They do.

Solution to Problem 2

We are required to find all prime numbers p such that $p^2 + 2$ is prime.

Checking what happens to the square numbers under division by 3 yields an easy answer to this problem.

We know that for integers n , if $n \equiv 0 \pmod{3}$, then $n^2 \equiv 0 \pmod{3}$, and if $n \equiv \pm 1 \pmod{3}$, then $n^2 \equiv 1 \pmod{3}$. In case you are not familiar with this fact, please verify it before proceeding further.

Suppose p is a prime number greater than 3. Then p is not divisible by 3, so $p \equiv \pm 1 \pmod{3}$, so $p^2 \equiv 1 \pmod{3}$, implying that $p^2 + 2$ is divisible by 3. Since $p^2 + 2$ is clearly greater than 3, this means that $p^2 + 2$ is not prime.

So $p^2 + 2$ is not a prime number for any prime number $p > 3$.

That leaves only these two primes to check: 2 and 3. The condition clearly does not hold for $p = 2$, and it does hold for $p = 3$, since $3^2 + 2 = 11$.

So the answer is that there is just one such prime number p , namely, $p = 3$.

Note: Problem 2 and the solution given above are based on Alon Amit's response to a Quora question (see <https://tinyurl.com/y529s4dm>). He points out that mathematicians believe that $n^2 + 2$ is prime for infinitely many values of n , but they do not know whether this is really so.

Solution to Problem 3

We are required to find all positive integers n such that $n^4 + 4$ is prime.

A problem of this kind invites us find a factorization of the given expression, $n^4 + 4$. This sub-problem yields to a familiar algebraic device: the

difference-of-two-squares factorization. We have,

$$\begin{aligned} n^4 + 4 &= n^4 + 4n^2 + 4 - 4n^2 \\ &= (n^2 + 2)^2 - (2n)^2 \\ &= (n^2 - 2n + 2)(n^2 + 2n + 2). \end{aligned}$$

This yields an explicit factorization of $n^4 + 4$.

Hence, if $n^4 + 4$ is prime, then the smaller of $n^2 - 2n + 2$ and $n^2 + 2n + 2$ must be ± 1 . The smaller number is clearly $n^2 - 2n + 2$. Hence,

$$n^4 + 4 \text{ prime} \implies n^2 - 2n + 2 = \pm 1.$$

The equation $n^2 - 2n + 2 = -1$ has no real solution (please check).

Hence $n^2 - 2n + 2 = 1$, giving $n^2 - 2n + 1 = 0$ and hence $n = 1$.

So if $n^4 + 4$ is prime, then n can only be 1.

As it happens, $n^4 + 4$ is prime when $n = 1$. So we have our answer: $n = 1$.

Solution to Problem 4

We are required to find the largest prime divisor p of $\binom{2000}{1000}$. It should be evident that we will not need to compute $\binom{2000}{1000}$ in full to find the answer; indeed, computing this number would be quite impractical as it has 601 digits!

Let us start with the standard expression for the given number,

$$\binom{2000}{1000} = \frac{2000!}{1000!1000!} = \frac{2000}{1} \times \frac{1999}{2} \times \frac{1998}{3} \times \cdots \times \frac{1001}{1000}.$$

From this it should be clear that all prime numbers between 1001 and 2000 are prime divisors of $\left(\frac{2000}{1000}\right)$, for there is nothing in the denominator with which they get divided away.

So, all we need to do is find the largest prime number between 1001 and 2000. To do so, we start with 2000 and go down 1 unit at a time, testing each number for primality. The first prime number we encounter will be the required answer.

Well, we are in luck because 1999 is itself a prime number! You may want to accept this as a known fact. Or you may want to look up tables of prime numbers; e.g., see https://www.walter-fendt.de/html5/men/primenumbers_en.htm. Check that the largest five prime numbers below 2000 are (in descending order) 1999, 1997, 1987, 1979, 1949. But if you want to demonstrate by yourself that 1999 is prime, here are the steps.

- First: compute $\left\lfloor \sqrt{1999} \right\rfloor = 44$, since $44^2 = 1936$ and $45^2 = 2025$.
- Next: list the prime numbers not exceeding 44. Here they are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43. We must check whether any of these is a divisor of 1999.
- Next: check mentally that none of 2, 3, 5, 7, 11 is a divisor of 1999.
- Next: check that 13 is not a divisor of 1999. Proof: $13 \times 15 = 195$, so 1950 is a multiple of 13, so we examine $1999 - 1950 = 49$; but this is clearly not a multiple of 13.
- Next: check that 17 is not a divisor of 1999. For proof we consider $1999 - 1700 = 299$. This number is not a multiple of 17 since 289 is a multiple of 17.
- Next: check that 19 is not a divisor of 1999. For proof we consider

the number $1999 - 1900 = 99$. This is not a multiple of 19, so neither is 1999.

- Proceeding in this manner we may verify that none of 23, 29, 31, 37, 41, 43 is a divisor of 1999. Reader, please do not accept our word for this; check it for yourself!
- We conclude that 1999 is a prime number .

Hence the required answer is 1999.

Solution to Problem 5

We must show that the integers $n^5 + n^4 + 1$ and $n^{101} + n^{37} + 1$ are not prime for any integer $n > 1$.

Both parts are resolved using the same idea. We seek factorizations of the polynomials $n^5 + n^4 + 1$ and $n^{101} + n^{37} + 1$. In both cases we use the non-real cube roots of unity, which are generally denoted by w and w^2 . They are the roots of the quadratic equation

$$z^2 + z + 1 = 0.$$

First we check that the equation does not have real roots (its discriminant is $1^2 - 4 = -3 < 0$). The two non-real roots are

$$\frac{-1 \pm i\sqrt{3}}{2}$$

where $i = \sqrt{-1}$. (Which root we call w and which one we call w^2 has no consequence.) Note that their product is 1 (i.e., $w^3 = 1$), and their sum is -1 (i.e., $w + w^2 = -1$ or $w^2 + w + 1 = 0$).

Let $f(z) = z^5 + z^4 + 1$. Then we have,

$$\begin{aligned} f(w) &= w^5 + w^4 + 1 \\ &= w^2 + w + 1 \quad (\text{because } w^3 = 1) \\ &= 0. \end{aligned}$$

Similarly we find that $f(w^2) = 0$. Hence w and w^2 are roots of $z^5 + z^4 + 1$. And since

$$(z - w)(z - w^2) = z^2 + z + 1,$$

it follows that $z^2 + z + 1$ is a factor of $z^5 + z^4 + 1$. This may be checked by actual division; we find that

$$z^5 + z^4 + 1 = (z^2 + z + 1) \cdot (z^3 - z + 1).$$

It follows that the integer $n^5 + n^4 + 1$ has $n^2 + n + 1$ and $n^3 - n + 1$ as divisors. It is easy to check for any $n > 1$, both divisors are greater than 1 and are therefore proper divisors. Hence $n^5 + n^4 + 1$ is composite for all $n > 1$. The only prime value taken by $n^5 + n^4 + 1$ occurs when $n = 1$.

The same argument works for $n^{101} + n^{37} + 1$ and we will not repeat it.

Solution to Problem 6

We are required to find all natural numbers n with the following property:
“Every natural number less than n can be expressed in exactly one way as the sum of distinct divisors of n .”

It is clear that all numbers n of the form 2^k , where k is a positive integer, have the required property. This is because the divisors of 2^k are the numbers $1, 2, 2^2, \dots, 2^{k-1}$, and it is well-known that every positive integer

can be expressed as a sum of distinct powers of 2 (this is the basis of the binary number system).

Are there any other positive integers with this property? We shall show that the answer is No, using the conditions described in the problem.

For example, consider the number 5. Its only proper divisor is 1, and this means that 2,3,4 cannot be expressed as sums of distinct divisors of 5.

More generally, no odd prime p has the required property because the number $p - 1$ cannot be expressed as a sum of distinct divisors of p .

Next, consider the number 6 whose proper divisors are 1,2,3. Observe that 3 can be expressed both as $1 + 2$ and as 3 itself. This violates the requirement of uniqueness.

More generally, we note that a number n that is not an odd prime and not a power of 2 must contain an odd prime divisor p that is strictly less than n . Then p can be expressed as a sum of distinct powers of 2 (from its binary representation; these powers are all less than p and hence less than n) and also as p itself. This immediately violates the requirement of uniqueness.

For some positive integers n , the other condition too may be violated: that every natural number less than n can be expressed as the sum of distinct divisors of n . For example, consider $n = 10 = 5 \cdot 2^1$. We may easily verify that 4 (which is less than 10) cannot be expressed as the sum of distinct divisors of 10.

More generally, let p be the smallest odd prime divisor of an even number n . Let $n = pm \cdot 2^k$ where $m \geq 1$ and $k \geq 1$. Suppose that $p > 2^{k+1}$. Consider the number 2^{k+1} . This number is less than n , since

$$n = pm \cdot 2^k \geq p \cdot 2^k > 2 \cdot 2^k = 2^{k+1}.$$

We claim that 2^{k+1} *cannot* be expressed as the sum of distinct divisors of n . For, to express 2^{k+1} in the required form, the only available divisors are $1, 2, 2^2, \dots, 2^k$, and their sum is $2^{k+1} - 1 < 2^{k+1}$.

Note: Problem 6 and the solution given above are based on Alex Eustis's response to a Quora question (see <https://tinyurl.com/3pznmrss>).

Kohli's Number 2997: Proofs

Article contributed by Hans Humenberger

In 2020 the Indian grade 12 student U. Kohli published an article titled "The Mystical Number 2997 (Kohli's Number)" see [1]. There he discovered a curious property of the natural number 2997. The paper does not give proofs and not everything is correct there (see footnote below), but nevertheless U. Kohli did a great job by discovering this curious property of the number 2997.

Take an arbitrary 3-digit number, say $N_0 = 257$, and iterate the following algorithm:

Multiply each digit of the number by 111 and add up all the resulting products.

Result: $2 \cdot 111 + 5 \cdot 111 + 7 \cdot 111 = 222 + 555 + 777 = 1554$

We denote this number by N_1 ; if the above algorithm is considered as a function f , one can say $N_1 = f(N_0)$.

The same procedure is applied to this result N_1 :

$$N_2 = f(N_1) = 1 \cdot 111 + 5 \cdot 111 + 5 \cdot 111 + 4 \cdot 111 = 111 + 555 + 555 + 444 = 1665$$

$$\text{Then a third time: } N_3 = f(N_2) = 1 \cdot 111 + 6 \cdot 111 + 6 \cdot 111 + 5 \cdot 111 = 111 + 666 + 666 + 555 = 1998$$

$$\text{And one more time: } N_4 = f(N_3) = 1 \cdot 111 + 9 \cdot 111 + 9 \cdot 111 + 8 \cdot 111 = 111 + 999 + 999 + 888 = 2997$$

And when one keeps proceeding in the same way one stays at 2997: $N_i = 2997$ for all $i \geq 4$.

Another example Let us take another 4-digit number, say $N_0 = 5762$. Then we have, step by step:

$$N_1 = f(N_0) = 5 \cdot 111 + 7 \cdot 111 + 6 \cdot 111 + 2 \cdot 111 = (5 + 7 + 6 + 2) \cdot 111 = 20 \cdot 111 = 2220,$$

$$N_2 = f(N_1) = 2 \cdot 111 + 2 \cdot 111 + 2 \cdot 111 + 0 \cdot 111 = (2 + 2 + 2) \cdot 111 = 6 \cdot 111 = 666,$$

$$N_3 = f(N_2) = 6 \cdot 111 + 6 \cdot 111 + 6 \cdot 111 = (6 + 6 + 6) \cdot 111 = 18 \cdot 111 = 1998,$$

and finally, again

$$N_4 = f(N_3) = 1 \cdot 111 + 9 \cdot 111 + 9 \cdot 111 + 8 \cdot 111 = (1 + 9 + 9 + 8) \cdot 111 = 27 \cdot 111 = 2997.$$

Obviously, there is one natural number that needs 0 steps to end up at 2997, namely 2997 itself. There are a few natural numbers that need only one step, like 999, 1998, or 3996.

Now, some questions arise: Does every positive natural number end up at 2997 under the iteration? How many steps do the numbers need at most to arrive at 2997?

Of course, based on computer experiments one could easily check a wide range of numbers experimentally, say all 3-digit numbers or all 4-digit numbers. But having done that and confirmed the phenomenon, we are left with at least two further questions:

1. Is there an explanation for the phenomenon? Why does the phenomenon hold?
2. What about numbers beyond the checked values?

Numbers with at Most Three Digits

Let us start to investigate the situation of numbers with at most 3 digits and prove that the iteration of f will always end up at 2997 after at most 4 steps.

As seen above, we can write f in another way: $f(N) = 111 \cdot DS(N)$, where DS denotes the digit sum. Because of the given start condition (at most 3-digits) we know $DS(N_0) \leq 27$.

From

$$N_1 = 111 \cdot DS(N_0)$$

we know that $N_1 \leq 111 \cdot 27 = 2997$.

What can be said about $DS(N_1)$? There are only a few numbers ≤ 2997 with digit sum > 27 , namely 1999, 2899, 2989. But these numbers cannot

serve as N_1 because they are not divisible by 111 (and not even by 3). Therefore, we can say:

$$DS(N_0) \leq 27 \Rightarrow DS(N_1) \leq 27,$$

analogously in all further steps: All N_i stay ≤ 2997 and all $DS(N_i)$ stay ≤ 27 .

Furthermore, the following hold:

- $3 \mid N_1$ (independent of N_0), because

$$N_1 = 111 \cdot DS(N_0).$$

- $3 \mid N_i \Rightarrow 9 \mid N_{i+1}$, because

$$N_{i+1} = 111 \cdot DS(N_i)$$

and from $3 \mid N_i$ we can conclude $3 \mid DS(N_i)$ (recall that a natural number is divisible by 3 if, and only if, its digit sum is divisible by 3), thus both factors in

$$N_{i+1} = 111 \cdot DS(N_i)$$

are divisible by 3, making N_{i+1} divisible by 9.

- Similarly, $9 \mid N_i \Rightarrow 27 \mid N_{i+1}$, because

$$N_{i+1} = 111 \cdot DS(N_i)$$

and the divisibility rule for 9, namely $9 \mid N_i \Leftrightarrow 9 \mid DS(N_i)$.

That means that if N_0 is not a multiple of 3, then every step (transition) in

$$N_0 \rightarrow N_1 \rightarrow N_2 \rightarrow N_3$$

produces one more factor 3 in the number. Therefore, N_3 is surely divisible by 27 and of course by 111, hence by 999. This leaves only three possibilities for N_3 , namely $N_3 = 999, 1998, 2997$ and in the first two cases the next step yields 2997. If N_0 itself is divisible by 3 or by 9, one ends up at 2997 even earlier. This explains why all natural numbers with not more than three digits end up at 2997 within at most four steps.

We could next investigate 4-digit numbers and 5-digit numbers. But we now take a really big step to googolplex $= 10^{10^{100}}$, the biggest number which has a “name” (we could even go further).

Numbers Bounded by Googolplex

It is not difficult to see that at most four iterations suffice even for really huge numbers, namely for starting numbers N_0 with at most

$$100 \dots 100 \dots 100 \quad (\text{with } 3 \cdot 10^{99} \text{ repetitions of '100'})$$

digits, a number with $9 \cdot 10^{99} \approx 10^{100}$ digits, i.e. roughly

$$\text{googolplex} = 10^{10^{100}}.$$

(*Comment.* Such big numbers do not play a role in any realistic context, since the number of elementary particles in the universe does not exceed 10^{100} .)

Then we have

$$DS(N_0) \leq 900 \dots 900 \dots 900 \quad (\text{with } 3 \cdot 10^{99} \text{ repetitions of '900'}),$$

and it follows that

$$\begin{aligned} N_1 &= 111 \cdot DS(N_0) \leq 111 \cdot 900 \dots 900 \dots 900 \\ &= 999999 \dots 999900 \quad (\text{with } 9 \cdot 10^{99} \text{ digits } 9, \text{ then } 00), \end{aligned}$$

which means that N_1 has at most $9 \cdot 10^{99} + 2$ digits, yielding

$$DS(N_1) < (9 \cdot 10^{99} + 2) \cdot 9 = 81 \cdot 10^{99} + 18.$$

Similarly we have

$$\begin{aligned} N_2 &= 111 \cdot DS(N_1) < 111 \cdot (81 \cdot 10^{99} + 18) \\ &= 8991 \cdot 10^{99} + 1998 < 10^{103}, \end{aligned}$$

telling us that N_2 has at most 103 digits and that

$$DS(N_2) < 103 \cdot 9 = 927.$$

We also know that $9 \mid N_2$ which implies $9 \mid DS(N_2)$, so

$$DS(N_2) \in \{9, 18, 27, \dots, 918\}.$$

Hence, for N_3 (after multiplication of $DS(N_2)$ by 111) the only possibilities for N_3 are:

$$999, \quad 1998, \quad 2997, \quad 3996, \quad \dots, \quad 101898,$$

the last number in the list being 999×102 . As all these numbers have $DS = 27$, it follows that $N_4 = 2997$.

This argumentation, exactly in the same way, could be extended to far beyond

$$\text{googolplex} = 10^{10^{100}},$$

in particular, for numbers N_0 with at most these many digits:

$$100 \dots 100 \dots 100 \quad (\text{with } 33 \cdot 10^{996} \text{ repetitions of '100'})$$

A number of this magnitude is approximately

$$10^{10^{998}};$$

here, it is crucial that all the multiples $999 \cdot k$ for $k = 1, \dots, 1000$ do have digit sum 27; see [2].

Thus, we see at this stage that the smallest natural number which needs more than four steps to end up at 2997 must be a really huge number.

Theorem. *Under the f -iteration all natural numbers eventually yield 2997.*

Applying f to a number n can have different effects: we may have $f(n) > n$ (as is the case for numbers with 1 or 2 digits), or $f(n) = n$ (as for 2997), or $f(n) < n$. Most often it is the third possibility that happens. We will now prove:

Theorem. *For all integers n with at least 5 digits, $f(n) < n$ (most of them drastically smaller).*

This means that they end up eventually in the above-mentioned region (numbers with at most these many digits:

$$100 \dots 100 \dots 100 \quad (\text{with } 3 \cdot 10^{99} \text{ repetitions of '100'}).$$

For this region we have already shown that 4 steps suffice to come to 2997.

Now for the proof that a number with at least 5 digits necessarily gets smaller under f .

Let N be a positive integer with n digits ($n \geq 5$), then we have

$$10^{n-1} \leq N < 10^n$$

and $DS(N) \leq 9 \cdot n$. Applying f yields

$$f(N) = 111 \cdot DS(N) \leq 999 \cdot n < 10^3 \cdot n.$$

Therefore,

$$\frac{f(N)}{N} < \frac{10^3 \cdot n}{N} \leq \frac{10^3 \cdot n}{10^{n-1}} = \frac{n}{10^{n-4}}$$

and this is < 1 for all $n \geq 5$ (if wanted, the fact $n < 10^{n-4}$ for $n \geq 5$ could be formally proved by complete induction; e. g. for $n = 5$ we have $5 < 10^1$ and for $n = 6$ we have $6 < 10^2$, with every step $n \rightarrow n + 1$ the left-hand side grows by $+1$ and the right-hand side by factor 10).

As stated above, the first number which needs more than 4 steps to converge to 2997 must be extraordinarily huge, but there are such numbers. It is challenging to find the smallest number needing 5 steps (or even 6 steps), but with some effort it is possible to construct a number that needs 5 steps. This we will do below.

A Number Which Needs 5 Steps

We construct such a number step by step, working backwards. We start at $N_4 = 3996$, because then we have $N_5 = 2997$, as wanted. Now we can write $N_4 = 3996 = 111 \cdot DS(N_3)$, i.e. we need an integer N_3 with $DS(N_3) = 36$ and $999 \mid N_3$ (note that every number N_3 must be divisible by 27 and 111, hence divisible by 999, see above). The smallest such number would be $999 \cdot 1011 = 1009989$, but this choice eventually causes problems with divisibility, therefore we take $N_3 = 999 \cdot 1071 = 1069929$.

Now we need an integer N_2 with

$$N_3 = 999 \cdot 1071 = 111 \cdot DS(N_2),$$

thus we are looking for an integer N_2 with $DS(N_2) = 9 \cdot 1071$ and $333 \mid N_2$ (note that every number N_2 must be divisible by 9 and 111, hence divisible by 333), so we take

$$N_2 = 99 \dots 99 \quad (\text{with } 1071 \text{ digits '9'}),$$

i.e., $N_2 = 10^{1071} - 1$. This number is divisible by 999, because one can divide these 1071 digits into 357 full '999' blocks.

Finally, we need a number N_1 with $N_2 = 111 \cdot DS(N_1)$, e.g. a number consisting of

$$\frac{N_2}{9 \cdot 111}$$

digits 9, thus we can write

$$N_1 = 10^{N_2/999} - 1 \approx 10^{10^{1068}},$$

a number with approximately 10^{1068} digits.

Here it is crucial that the number of digits 9 in N_1 (i.e., $N_2/999$) is divisible by 3, so that N_1 can be split in full '999' blocks which guarantees $111 \mid N_1$ and also $999 \mid N_1$. This would have not been the case if we had taken the factor 1011 instead of 1071 in the construction of N_3 ; then $N_2/999$ would be a number consisting of 337 digits 1 and many 0's, i.e. not divisible by 3. Because of $999 \mid N_1$, the next step works the same way:

$$N_0 = 10^{N_1/999} - 1 \approx 10^{10^{1068}},$$

a number with approximately 10^{1068} digits.

Remark

If $N_1/999$ (the number of digits 9 in N_0) were divisible by 3 (this is *not* the case here, unfortunately; the number $N_2/999$ of digits 9 in N_1 is given by $N_2/999 = "357 \text{ blocks of the form } 001"$, i.e., this number is divisible by 3 but not by 9; $N_2/(3 \cdot 999)$ is the number of full "999" blocks in N_1 , that means $N_1/999$ has $N_2/(3 \cdot 999)$ many blocks of the form "001".

Finally, note the equivalences

$$3 \mid \frac{N_1}{999} \Leftrightarrow 3 \mid \frac{N_2}{3 \cdot 999} \Leftrightarrow 9 \mid \frac{N_2}{999},$$

but 357 is *not* divisible by 9) one could continue similarly and construct a number which needs 6 steps to end up at 2997, i.e., $10^{N_0/999} - 1$. If we had taken the factor 1161 instead of 1071 above in the construction of N_3 , then with $10^{N_0/999} - 1$ we would indeed have a number which needs 6 steps!

For using the above principle generally, especially in the direction " $\Leftarrow$ ", the following (easy to prove) result is important:

$$3 \mid m \Rightarrow 999 \mid (10^m - 1).$$

For a proof, let $m = 3k$, then $10^m - 1 = 10^{3k} - 1$ is a number consisting of $3k$ digits 9, which can be grouped in k blocks of '999',

$$999999 \dots 999,$$

and this number is divisible by 999, the quotient being given by

$$1001001001, \dots 001.$$

Further reading

- [1] Kohli, U. (2020). "The Mystical Number 2997 (Kohli's Number)". *At Right Angles* (8). pp. 35-36. <https://publications.azimpremjiuniversity.edu.in/2588/>
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About the author

Hans Humenberger received his PhD at the University of Vienna in the field of mathematics education. He works at the same university as a professor of mathematics and mathematics education. He also heads the working group *Didactics of mathematics and school mathematics* which is responsible for the education of teachers at secondary and high school level. He has written many papers and books in mathematics education and mathematics. His main fields of interest are mathematics as a process, applications of mathematics, problem solving, geometry and stochastics; see <https://homepage.univie.ac.at/hans.humenberger/>. He may be contacted at hans.humenberger@univie.ac.at. His mailing address: University of Vienna, Faculty for Mathematics, Oskar-Morgenstern-Platz 1, 1090 Vienna, Austria.

Making Friends With Friendly Numbers

Article contributed by Pustak Pathak

Introduction

Numbers are friends of human beings, but some people have deeper friendships with numbers. They see patterns in simple-looking numbers. One such person was the great Indian mathematician Srinivasa Ramanujan. There are many other examples, and there are even novels about such people. Recently I read one such by Yoko Ogawa [1]; it is about a professor who has partial memory loss and who can only remember things from the last 80 minutes. He is fascinated by certain numbers. In this article we talk about such numbers. (For a review of the novel, see [2].)

We mentioned that numbers are our friends. Can numbers also be friends of each other? Yes, there are “friendly numbers”. We describe below what this means.

Notation: $s(n)$ denotes the sum of all the *proper divisors* of n , and $\sigma(n)$ denotes the sum of *all the divisors* of n . E.g., $s(10) = 8$ and $\sigma(10) = 18$.

Some families of numbers

Amicable pairs: An *amicable pair* is a pair $\{m, n\}$ of natural numbers, each of which is the sum of the proper divisors of the other, i.e., $s(m) = n$ and $s(n) = m$. For example, $\{220, 284\}$. The proper divisors of 220 are 1, 2, 4, 5, 10, 11, 20, 22, 44, 55, 110, and their sum is

$$s(220) = 1 + 2 + 4 + 5 + 10 + 11 + 20 + 22 + 44 + 55 + 110 = 284.$$

The proper divisors of 284 are 1, 2, 4, 71, 142, and their sum is

$$s(284) = 1 + 2 + 4 + 71 + 142 = 220.$$

Such pairs are rare. The smallest such pair, known to the Greeks, is the one noted above, $\{220, 284\}$. It was the only pair known till the 14th century when $\{17296, 18416\}$ was found.

Then in the 17th century, the pair $\{9363584, 9437056\}$ was found. Till the end of the 17th century, mathematicians knew of only three such pairs.

Then in the 18th century, Euler discovered 59 amicable pairs, and in 1866, an Italian student Nicolò Paganini found the second smallest amicable pair, $\{1184, 1210\}$, which had been missed by everyone else till then.

Perfect numbers: If a natural number n satisfies the relation $s(n) = n$, it is called a *perfect number*. Examples: 6 and 28 (check: $s(6) = 1 + 2 + 3 = 6$ and $s(28) = 1 + 2 + 4 + 7 + 14 = 28$). To date we know of only 52 perfect numbers; the last one was found in 2024. All currently known perfect numbers are even; no odd perfect number is known.

Abundant numbers: If a natural number n satisfies the relation $s(n) > n$, it is called an *abundant number*. The smallest abundant number is 12 (check: $s(12) = 1 + 2 + 3 + 4 + 6 = 16 > 12$).

Similarly, $s(18) = 1 + 2 + 3 + 6 + 9 = 21 > 18$, so 18 is abundant.

There are 21 abundant numbers below 100, and all are even, but this does not mean that odd abundant numbers do not exist. The smallest odd abundant number is 945.

For perfect numbers and amicable pairs, the question that whether their number is infinite is open, but for abundant numbers, we can easily show that their number is infinite. This is so because a multiple of an abundant number is itself abundant. For, if m is abundant (i.e., $s(m) > m$) and $k > 1$ is any positive integer, then the divisors of km include all numbers of the form k times a divisor of m ; and there is also the divisor 1 (not included in the above); these give $s(km) \geq km + 1 > km$; hence km is abundant.

Deficient numbers: If a natural number n satisfies the relation $s(n) < n$, it is called a *deficient number*. All prime numbers are deficient, as they have just one proper divisor which is 1. Similarly, any prime power p^k where k is a positive integer greater than 1 is deficient since

$$s(p^k) = 1 + p + \cdots + p^{k-1} = \frac{p^k - 1}{p - 1} < p^k - 1 < p^k.$$

As the number of prime numbers is infinite, the number of deficient numbers is infinite as well.

Any natural number is perfect, or abundant, or deficient. Among these types, deficient numbers are more common. There are 21 abundant numbers and 2 perfect numbers below 100; the remaining are deficient.

Abundancy index: Let us define the *abundancy index* of n by $A = \sigma(n)/n$. Then, perfect numbers have $A = 2$, abundant numbers have $A > 2$, and deficient numbers have $A < 2$. This notation will come of use in defining ‘friendly’ numbers.

We note in passing that the sum of the reciprocals of the divisors of a perfect number is 2. E.g., 6 is a perfect number having 1, 2, 3, 6 as divisors, and $1/1 + 1/2 + 1/3 + 1/6 = 2$. Readers may try to prove this property.

Friendly numbers: We define *friendly numbers* to be a collection of two or more natural numbers with the same abundancy index A .

For perfect numbers, $A = 2$. Consequently, all perfect numbers are friends of each other.

We can have collections of natural numbers whose abundancy index is not 2. For example, consider $\{30, 140\}$. Since $\sigma(30) = 72$ and $\sigma(140) = 336$, both numbers have abundancy index $12/5$. These two numbers have other friends: 2480, 6200, 40640 all have abundancy index $12/5$. So these are examples of even numbers which are friendly and abundant.

An example of friendly odd, deficient numbers is the pair $\{135, 819\}$; they have abundancy index $A = 16/9 < 2$.

As mentioned earlier, interest in these numbers started early, but it took time to find many examples. With the availability of computers and some knowledge of programming, it becomes much easier to find such numbers.

How things have changed with the advent of modern computers

As noted above, till the advent of computers we knew only a handful of amicable pairs and perfect numbers. The source [3] tells us that we now know 1,227,869,886 amicable pairs, thanks to computers. A distributed computing project has found all pairs smaller than 10^{20} , whereas till the middle of 20th century, only 390 such pairs were known. But it is still not known whether the number of amicable pairs is infinite.

What we can do with our personal computers

Finding perfect numbers is difficult. Even with the help of modern computers, to date we know only 52 perfect numbers. The first two perfect numbers (6 and 28) are easily found, but beyond that it is difficult to find them manually. A small computer program can do so quickly. The author has written three easy-to-follow programs in Python. The code listings may be found at the following URL:

https://colab.research.google.com/drive/1t5Mifqe-HGwT_yhPFShfvWMWVrSvdMjb?usp=sharing

- Program 1 yields all perfect numbers up to any given number. The algorithm used is not very efficient. To find large perfect numbers requires a lot of time.
- Program 2 yields all perfect numbers up to any given number. It works faster than Program 1. The algorithm is much more efficient.
- Program 3 yields all pairs of amicable numbers in the given range.

The logic used in these programs can easily be modified to find abundant, deficient and friendly numbers. This is left to the reader.

It turns out that all known perfect numbers are even (see the table below) and end with 6 or 8. The same table also highlights the fact that all these perfect numbers have the form $2^{p-1} \times (2^p - 1)$ where both p and $2^p - 1$ are prime numbers.

Prime numbers of the form $2^p - 1$ are referred to as *Mersenne primes*. They form an important class of prime numbers.

6	=	2×3	=	$2^1(2^2 - 1)$
28	=	4×7	=	$2^2(2^3 - 1)$
496	=	16×31	=	$2^4(2^5 - 1)$
8128	=	64×127	=	$2^6(2^7 - 1)$
33550336	=	4096×8191	=	$2^{12}(2^{13} - 1)$

Table 1: Perfect numbers and their prime factorizations

It can be shown that there is a one-to-one correspondence between the Mersenne primes and even perfect numbers; each even perfect number can be associated with a Mersenne prime, and each Mersenne prime generates an even perfect number. Hence, the problem of finding even perfect numbers is essentially the same as the problem of finding Mersenne primes. The largest known perfect number, found very recently, is

$$2^{82589932} \times (2^{82589933} - 1);$$

it is associated with the Mersenne prime $2^{82589933} - 1$.

Regarding odd perfect numbers, extensive searches (based on advanced mathematics) have revealed that there do not exist any odd perfect numbers below 10^{1500} . Hence, if there exists an odd perfect number it is greater than 10^{1500} . But we do not know whether there exist any such numbers.

Conclusion

Numbers continue to fascinate us. For example, the author and his father routinely observe number plates of cars and motorbikes to find hidden patterns. Other number theory enthusiasts may have their own way to entertain themselves with numbers. We conclude this article with a hope that some readers will find that their date of birth, roll number, vehicle number, etc., are not just numbers, they have interesting properties.

Further reading

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- [2] Chari, N., 2022. “The Housekeeper and the Professor”, *At Right Angles*, (14), pp.74-76,
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About the author

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The Angle Bisector Theorem – Part 2

This is a continuation of the article begun in the last issue dealing with the angle bisector theorem. We consider a few applications this time, mostly related to trigonometry. We conclude with a trigonometric solution of a famous problem in geometry.

Finding $\tan 15^\circ$

We consider a right-angled triangle ABC in which $\angle C = 90^\circ$ and $\angle B = 30^\circ$ (see Figure 7). Let $AB = 2$; then

$$AC = 2 \sin 30^\circ = 1,$$

$$BC = 2 \cos 30^\circ = \sqrt{3}.$$

Let the internal bisector of $\angle ABC$ meet AC at D , so $\angle DBC = 15^\circ$.

Let $DC = x$, $AD = 1 - x$.

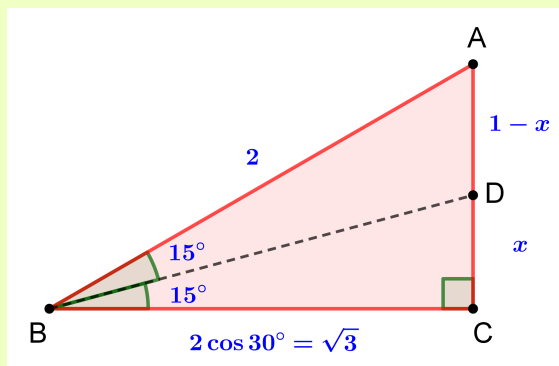


Figure 7: Finding $\tan 15^\circ$ using the angle bisector theorem

The angle bisector theorem tells us that

$$\frac{x}{1-x} = \frac{BC}{BA} = \frac{\sqrt{3}}{2},$$

so $2x = \sqrt{3} - x\sqrt{3}$, giving

$$x = \frac{\sqrt{3}}{2 + \sqrt{3}} = \sqrt{3}(2 - \sqrt{3}).$$

Hence

$$\tan 15^\circ = \frac{x}{\sqrt{3}} = 2 - \sqrt{3}. \quad (1)$$

There are several other ways of obtaining the same result. We invite the reader to explore these possibilities. ■

Finding $\tan 22.5^\circ$

We consider a right-angled triangle ABC in which $\angle C = 90^\circ$ and $\angle B = 45^\circ$ (see Figure 8). Let $BC = 1$; then

$$AC = 1, \quad AB = \sqrt{2}.$$

Let the internal bisector of $\angle ABC$ meet AC at D , so $\angle DBC = 22.5^\circ$.

Let $DC = x$, $AD = 1 - x$.

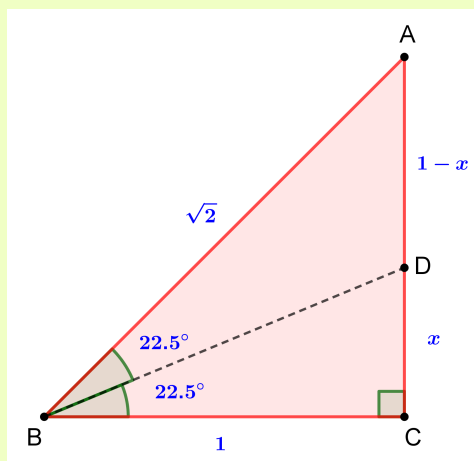


Figure 8: Finding $\tan 22.5^\circ$ using the angle bisector theorem

The angle bisector theorem tells us that

$$\frac{x}{1-x} = \frac{BC}{BA} = \frac{1}{\sqrt{2}},$$

so $x\sqrt{2} = 1 - x$, giving

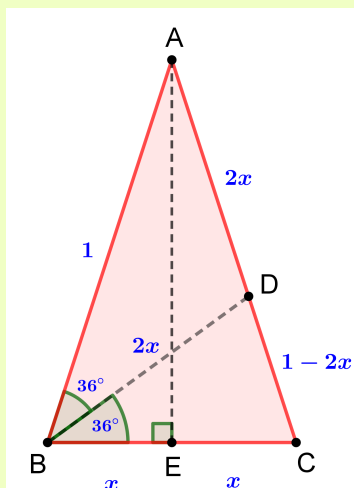
$$x = \frac{1}{\sqrt{2} + 1} = \sqrt{2} - 1.$$

Hence

$$\tan 22.5^\circ = \frac{x}{1} = \sqrt{2} - 1. \quad (2)$$

As earlier, there are other ways of obtaining the same result. ■

Finding $\cos 72^\circ$



We consider an isosceles triangle ABC in which $\angle B = 72^\circ = \angle C$ and $\angle A = 36^\circ$ (see Figure 9). Let $AB = 1$; then $AC = 1$.

Let the internal bisector of $\angle ABC$ meet AC at D , so $\angle DBC = 36^\circ$.

Since $\angle DCB = 72^\circ$, it follows that $\angle BDC = 72^\circ$; hence $BD = BC$.

Since $\angle ABD = 36^\circ = \angle BAD$, we also have $AD = BD$. This implies that $BC = BD = AD$.

Figure 9: Finding $\cos 72^\circ$ using the angle bisector theorem

Let E be the midpoint of BC ; join AE . Then $AE \perp BC$.

Let $BE = x$; then $EC = x$, $BC = 2x$, $BD = 2x$, $AD = 2x$, so $DC = 1 - 2x$.

The angle bisector theorem tells us that

$$\frac{1 - 2x}{2x} = \frac{BC}{BA} = \frac{2x}{1}.$$

Hence $1 - 2x = 4x^2$, i.e., $4x^2 + 2x - 1 = 0$. We solve this quadratic equation:

$$x = \frac{-2 \pm \sqrt{4 + 16}}{8} = \frac{-1 \pm \sqrt{5}}{4}.$$

The negative sign clearly does not apply here (since $x > 0$); hence

$$x = \frac{\sqrt{5} - 1}{4}.$$

A glance at Figure 9 tells us what this means; namely, that

$$\cos 72^\circ = \frac{\sqrt{5} - 1}{4}. \quad (3)$$

This result too can be obtained in other ways.

Remark

The reader may be aware of the Golden Ratio Φ which is defined by

$$\Phi = \frac{\sqrt{5} + 1}{2} \approx 1.618.$$

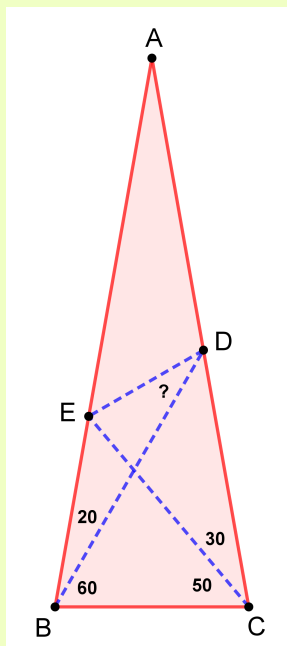
We see that in terms of the golden ratio we may express $\cos 72^\circ$ as follows:

$$\cos 72^\circ = \frac{\Phi - 1}{2}. \quad (4)$$

Because of the relationship between the Golden Ratio and the trigonometric ratios of 72° , the isosceles triangle with angles $36^\circ, 72^\circ, 72^\circ$ is known as the *Golden Triangle*.

Problem: The ‘80-80-20’ triangle

We conclude with an analysis of a famous problem concerning the isosceles triangle with angles $80^\circ, 80^\circ, 20^\circ$.



In the figure, triangle ABC is isosceles with $\angle ABC = 80^\circ = \angle ACB$. Points D on AC and E on AB are located such that

$$\angle DBC = 60^\circ, \quad \angle ECB = 50^\circ.$$

The problem is to find the size of $\angle BDE$.

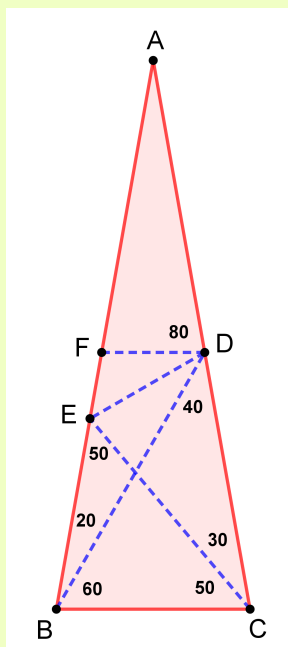
There are several elegant pure geometry solutions of the problem — but they are very challenging to find!

The problem was posed by Edward Mann Langley in 1922, in *The Mathematical Gazette*. Several wonderful solutions have appeared over the decades.

Figure 10: Langley’s Adventitious Angles problem

We opt for a solution using trigonometry. The solution makes use of the sine rule for triangles and the well-known identity

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}.$$



Construction: Draw DF parallel to CB , with F on AB . Note the sizes of a few angles (these are the 'obvious' ones):

$$\angle BEC = 50^\circ, \quad \angle BDC = 40^\circ,$$

$$\angle ADF = 80^\circ, \quad \angle BDF = 60^\circ.$$

We shall prove that DE bisects $\angle BDF$. This will imply that $\angle BDE = 30^\circ$.

We do this by using the converse of the angle bisector theorem. This means that we must first show that

$$\frac{DF}{DB} = \frac{FE}{EB}.$$

We shall do this computationally.

Figure 11: Solution to Langley's Adventitious Angles problem

Since $\angle BEC = 50^\circ = \angle BCE$, we have $BE = BC$. Also, $CD = BF$. Hence

$$FE = BF - BE = CD - BC,$$

and

$$\frac{FE}{EB} = \frac{CD - BC}{BC} = \frac{CD}{BC} - 1.$$

The sine rule (in $\triangle BCD$) tells us that

$$\frac{CD}{BC} = \frac{\sin 60^\circ}{\sin 40^\circ}.$$

Hence,

$$\frac{FE}{EB} = \frac{\sin 60^\circ}{\sin 40^\circ} - 1.$$

Using the identity noted above, we obtain,

$$\frac{\sin 60^\circ}{\sin 40^\circ} - 1 = \frac{\sin 60^\circ - \sin 40^\circ}{\sin 40^\circ} = \frac{2 \cos 50^\circ \sin 10^\circ}{\sin 40^\circ} = 2 \sin 10^\circ,$$

since $\cos 50^\circ = \sin 40^\circ$. Hence,

$$\frac{FE}{EB} = 2 \sin 10^\circ.$$

Next, invoking the sine rule in $\triangle BDF$, we get

$$\begin{aligned} \frac{DF}{DB} &= \frac{\sin 20^\circ}{\sin 100^\circ} = \frac{\sin 20^\circ}{\sin 80^\circ} \\ &= \frac{\sin 20^\circ}{\cos 10^\circ} = 2 \sin 10^\circ, \end{aligned}$$

since $\sin 20^\circ = 2 \sin 10^\circ \cos 10^\circ$.

It follows that

$$\frac{FE}{EB} = \frac{DF}{DB}.$$

By the converse of the angle bisector theorem we conclude that DE bisects $\angle BDE$. Hence $\angle BDE = 30^\circ$. ■

As noted earlier there are many extremely elegant solutions to this problem. We invite the reader to study them — and marvel at their ingenuity! See in particular [1] and [2].

Problems for the reader

Prove the following.

1. $\tan 7.5^\circ = \sqrt{6} - \sqrt{3} + \sqrt{2} - 2$
2. $\sin 22.5^\circ = \frac{1}{2} \sqrt{2 - \sqrt{2}}$
3. $\tan 11.25^\circ = \sqrt{(2\sqrt{2} + 4)} - \sqrt{2} - 1$
4. $\tan 36^\circ = \sqrt{5 - 2\sqrt{5}}$
5. $\tan 18^\circ = \frac{1}{\sqrt{5}} \sqrt{5 - 2\sqrt{5}}$

Further reading

- [1] CoMaC (Shailesh Shirali), “A 80-80-20 Triangle” from <https://publications.azimpremjiuniversity.edu.in/1440/1/13-A%2080-80-20%20Triangle.pdf>
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Gems From Geometry – 4

In this article we study a famous and important result in geometry called *Ptolemy's Theorem*. Here is its statement (see Figure 12).

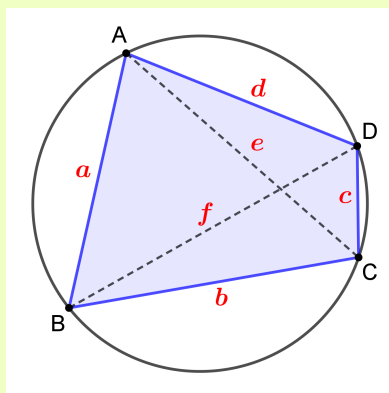


Figure 12: Ptolemy's theorem tells us that $ac + bd = ef$

Theorem (Ptolemy of Alexandria). *If ABCD is a cyclic quadrilateral, then*

$$AB \cdot CD + BC \cdot AD = AC \cdot BD. \quad (5)$$

In words: “*The sum of the products of opposite pairs of sides of a cyclic quadrilateral is equal to the product of the diagonals.*”

This result is named after Claudius Ptolemy (c. 100–170 CE), an influential Greek astronomer and mathematician. Ptolemy not only stated the theorem but also gave a proof in his famous astronomical work, the *Almagest*. He used this relationship to construct his table of chords (this is an early form of a trigonometric table which was crucial for astronomical calculations).

We shall make further remarks about the history of Ptolemy’s theorem in the next part of the article (in the next issue).

A ‘pure geometry’ proof

To prove the theorem presents a challenge. The difficulty lies in the fact that neither side of the equality $AB \cdot CD + AD \cdot BC = AC \cdot BD$ seems to *mean* anything. Terms like $AB \cdot CD$ and $AD \cdot BC$ suggest areas — but of what? There is nothing in the figure that yields a clue.

So we try a different approach. We write the equality to be proved as

$$\frac{AB \cdot CD}{BD} + \frac{BC \cdot AD}{BD} = AC. \quad (6)$$

Have we made progress by writing it this way? Perhaps. Now the equality to be proved is a relation between *lengths*. [Can we now find or construct two segments whose lengths together yield the length of AC?](#)

The expressions on the left ($AB \cdot CD/BD$ and $BC \cdot AD/BD$) suggest that we must look for, or construct, suitable pairs of similar triangles. Indeed, the form $AB \cdot CD/BD$ suggests that we should construct a triangle similar to $\triangle ABD$, and moreover that this triangle (which is yet to be constructed) should have CD for a side.

Noting that $\angle ABD = \angle ACD$ we ask: **What if we locate a point E on AC such that $\triangle ABD \sim \triangle ECD$?** Then we would have $AB/BD = EC/CD$, giving $EC = AB \cdot CD/BD$; just what we want! Now we have a clue on how to proceed. Figure 13 shows the construction.

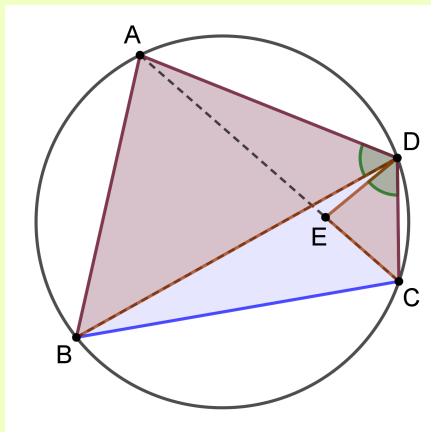


Figure 13: Construction of an appropriate point E on diagonal AC

Locate a point E on diagonal AC such that $\angle CDE = \angle ADB$ (the two angles are marked in Figure 13). Now consider $\triangle CDE$ and $\triangle ADB$. Since $\angle ECD = \angle ABD$ by the angle property of a circle, and $\angle CDE = \angle ADB$ by design, we have $\triangle CDE \sim \triangle ADB$. Hence:

$$\frac{EC}{CD} = \frac{AB}{BD}, \quad \therefore EC = \frac{AB \cdot CD}{BD}. \quad (7)$$

With reference to the same figure we also have $\triangle DAE \sim \triangle DBC$, because $\angle DAE = \angle DBC$ and $\angle ADE = \angle BDC$.

Hence:

$$\frac{BC}{BD} = \frac{AE}{AD}, \quad \therefore AE = \frac{BC \cdot AD}{BD}. \quad (8)$$

From equalities (7) and (8) we get, since $AE + EC = AC$:

$$AC = \frac{AB \cdot CD}{BD} + \frac{BC \cdot AD}{BD},$$

$$\therefore AC \cdot BD = AB \cdot CD + BC \cdot AD, \quad (9)$$

as was to be proved. ■

You will agree that this is an extremely elegant proof (it is the proof given by Ptolemy). It is something to be admired — but it would be a challenge to find it on one's own!

Ptolemy's theorem can be proved in other ways. Of particular interest are: (i) a proof using complex numbers, (ii) a proof using vectors, (iii) a proof based on a geometrical transformation called 'inversion'. We will study these ways in later articles.

Four applications of Ptolemy's theorem

Application to an equilateral triangle

Theorem. *Let ABC be an equilateral triangle, and let P be any point on the circumcircle of the triangle. Then the largest of the distances PA, PB, PC is equal to the sum of the other two smaller distances.*

The theorem is illustrated in Figure 14. Here P is located on the minor arc BC , i.e., it lies between the points B and C , making PA the largest among PA, PB, PC . The theorem now asserts that $PA = PB + PC$.

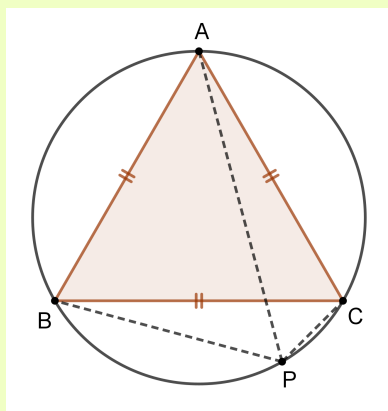


Figure 14: Ptolemy's theorem tells us us that $PA = PB + PC$

Proof

The proof is simplicity itself. Consider the cyclic quadrilateral $PBAC$. Apply Ptolemy's theorem to it; we get:

$$PA \cdot BC = PB \cdot AC + PC \cdot AB. \quad (10)$$

The equal factor ($AB = BC = CA$) present in the three terms in (10) cancels out, and we are left with the desired relation, $PA = PB + PC$. ■

Remark The theorem can also be proved using the following identity: for all angles θ (measured in degrees),

$$\sin(60 - \theta) + \sin \theta = \sin(60 + \theta). \quad (11)$$

We invite the reader to complete the proof.

Application to a regular pentagon

If you examine a regular pentagon, you will notice that its five diagonals all have the same length. (Try to justify why this is so.)

Theorem. *Given a regular pentagon with side 1, let its diagonals have length x . Then we have the following relation:*

$$x^2 = x + 1. \quad (12)$$

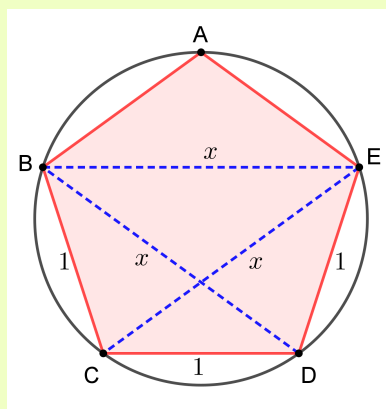


Figure 15: Application of Ptolemy's theorem to a regular pentagon

Proof

In Figure 15, $ABCDE$ is a regular pentagon; its sides have length 1 and its diagonals have length x (we have shown only three of the diagonals). We apply Ptolemy's theorem to the inscribed quadrilateral $BCDE$; we get:

$$BC \cdot DE + BE \cdot CD = BD \cdot CE. \quad (13)$$

That is, $1 + x = x^2$, as required. ■

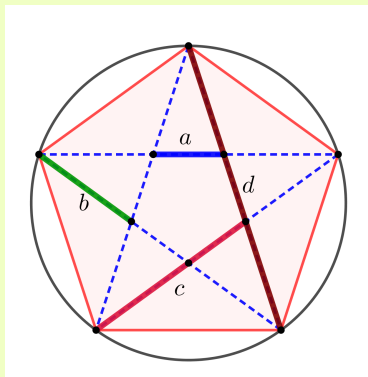
Solving the equation $x^2 = x + 1$ we get: $x = (1 \pm \sqrt{5})/2$. The negative sign cannot hold, since x is positive; hence we have:

$$x = \frac{\sqrt{5} + 1}{2}. \quad (14)$$

So the ratio of the diagonal to the side of a regular pentagon is φ , the Golden Ratio!

Problem

Before presenting the next application we share an interesting problem to do with a regular pentagon.



If we draw a regular pentagon and all its diagonals and study the configuration carefully, we find segments with four different lengths within the figure (with several copies of each one). We have coloured them differently here and marked them a, b, c, d (with $a < b < c < d$).

Figure 16: Lengths within a regular pentagon

Here comes the lovely finding. **The lengths of a, b, c, d are in geometric progression, the common ratio being the golden ratio, φ !**

Try showing this on your own.

Application: a trigonometric inequality

Our third result is a trigonometric inequality which would seem difficult to prove using purely geometric methods. We shall show that for any acute angle θ ,

$$\sin \theta + \cos \theta \leq \sqrt{2}. \quad (15)$$

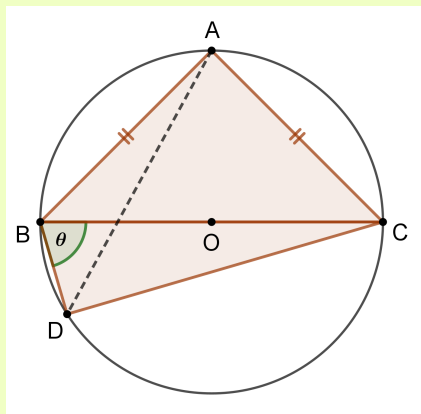


Figure 17 shows a circle with unit radius; BC is a diameter; A and D lie on the circle, on opposite sides of BC , with $AB = AC$; and $\angle DBC = \theta$.

$\triangle ABC$ is right-angled, isosceles; $\triangle DBC$ too is right-angled. Hence $AB = AC = \sqrt{2}$, $BD = 2 \cos \theta$, $CD = 2 \sin \theta$, $BC = 2$.

Figure 17: Application of Ptolemy's theorem to a trig inequality

We apply Ptolemy's theorem to quadrilateral $ABDC$:

$$AB \cdot CD + AC \cdot BD = AD \cdot BC,$$

$$\therefore 2\sqrt{2} \sin \theta + 2\sqrt{2} \cos \theta = 2AD.$$

As AD is a chord of the circle and so does not exceed in length the diameter

of the circle, $AD \leq 2$. This yields

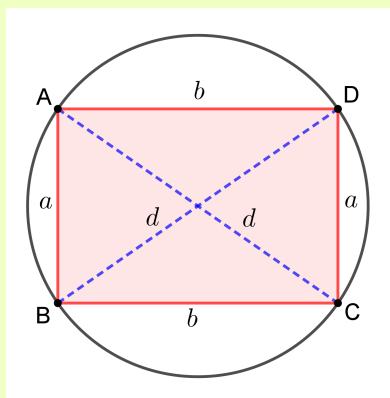
$$2\sqrt{2} \sin \theta + 2\sqrt{2} \cos \theta \leq 4,$$

$$\therefore \sin \theta + \cos \theta \leq \sqrt{2},$$

as claimed. ■

Application: Pythagoras theorem

To close this section, we describe a particularly simple application — a proof of the Pythagorean theorem.



Shown in the figure is a rectangle $ABCD$, its diagonals AC, BD , and its circumscribing circle. Let the sides of the rectangle be a, b , and let the two diagonals have length d . Since a rectangle is cyclic we can apply the theorem of Ptolemy. We get:

$$a^2 + b^2 = d^2,$$

which is the theorem of Pythagoras.

Figure 18: Proof of the Pythagorean theorem

In the next issue we shall showcase more applications of Ptolemy's theorem and show why the theorem holds great significance for the development of trigonometry. We shall also prove an inequality version of the theorem.

Later we shall explore whether Ptolemy's theorem was known and made use of by mathematicians in ancient India. This question is of great interest to historians of mathematics.

Further reading

- [1] A. Bogomolny, "Ptolemy's Theorem" from *Interactive Mathematics Miscellany and Puzzles*,
<http://www.cut-the-knot.org/proofs/ptolemy.shtml>
- [2] Wikipedia, "Ptolemy's Theorem" from
https://en.wikipedia.org/wiki/Ptolemy's_theorem
- [3] W. Derrick, J. Herstein (2012), "Proof Without Words: Ptolemy's Theorem", *The College Mathematics Journal*, v.43, n.5, p. 386

Article contributed by Shailesh Shirali

Did You Know ... ? – 2

The Beal conjecture

The reader will be familiar with the story of [Fermat's Last Theorem](#) (FLT for short). The story of FLT goes all the way back to 1637, when a bold assertion was made by Pierre de Fermat (a jurist and amateur mathematician) in the margin of a famous number theory book. Expressed in modern language, this is the claim that for integers $n > 2$, there are no positive integers a, b, c satisfying the equality

$$a^n + b^n = c^n.$$

Fermat died in 1665 and the statement that he had scribbled in the margin of a book came to light. (Alongside, he had made a second claim: that he had found a “truly remarkable proof” of the claim but that the margin was too small to hold it.) But no proof of the FLT was ever found in his writings, and the challenge of proving (or disproving) the claim caught the attention of many mathematicians.

Over the next three and a half centuries, many ‘proofs’ were found but they were all found wanting: they invariably contained subtle flaws (though

just to spot those flaws required deep insight). There are probably more than a hundred such faulty proofs!

Ultimately, mathematicians triumphed. In 1993-94 a British born Princeton mathematician named Andrew Wiles found a proof (with assistance from his student Richard Taylor), and that proof was accepted by experts — but only after extremely close scrutiny! We will not say anything further about this extraordinary story, as it has been narrated in many excellent articles. See [1], [2], [3], and [4].

The reason we have written about FLT is that in 1993, a related conjecture came up which closely resembles FLT. It is known as the [Beal conjecture](#). Here is its statement (from [5])

Conjecture (Andrew Beal, 1993). *If $A^x + B^y = C^z$ where A, B, C, x, y , and z are positive integers with $x, y, z > 2$, then A, B , and C have a common prime factor.*

Or, equivalently:

Conjecture (Andrew Beal, 1993). *The equation $A^x + B^y = C^z$ has no solutions in positive integers and pairwise coprime integers A, B, C if $x, y, z > 2$.*

It is relatively easy to find positive integers A, B, C and integers $x, y, z > 2$ such that $A^x + B^y = C^z$. But in all these instances we find that A, B, C are not pairwise coprime. For example,

$$3^3 + 6^3 = 3^5 \quad (\text{common factor } 3),$$

$$7^3 + 7^4 = 14^3 \quad (\text{common factor } 7),$$

$$2^n + 2^n = 2^{n+1} \quad (\text{common factor } 2).$$

We invite the reader to generate more such instances. It is a pleasant exercise.

Connection between FLT and the Beal conjecture

Is there a connection between the two? Yes. *FLT is a special case of the Beal conjecture.* So if anyone proves the Beal conjecture, it immediately implies that FLT is true. (But the truth of FLT does not imply the truth of the Beal conjecture.) Let us justify this claim.

Theorem. *If the Beal conjecture is true, then FLT is true.*

We shall prove this in the contrapositive form:

Theorem. *If FLT is false, then the Beal conjecture is false.*

Proof Suppose that a solution exists to FLT; i.e., suppose that two positive n -th powers exist that add up to another positive n -th power, with $n > 2$.

By dividing out all common factors, we would then obtain a solution to $A^n + B^n = C^n$ with A , B , and C coprime. But this directly contradicts the Beal conjecture (in the second form).

It follows that FLT is a special case of the Beal conjecture with $x = y = z$. ■

Reward!

The conjecture was first stated in 1993 by Andrew Beal, a millionaire banker and amateur mathematician. Later he offered prize money for a proof of the conjecture or a counterexample. Currently, there is a price of \$1 million waiting for anyone who settle the conjecture one way or the other!

But, given the fact that the Beal conjecture if true instantly implies the truth of FLT, no one expects a quick and easy answer to the challenge!

In general, one may wonder what makes the Beal conjecture so difficult to prove or disprove. Alon Amit writes (see <https://www.quora.com/What-makes-the-Beal-conjecture-so-difficult-to-prove-or-disprove/answer/Alon-Amit>), “Put simply, equations of high degree with several variables that are to be solved in integers are very, very difficult. Why? Well, they just are. We’re often not smart enough to solve them.” He adds “if you’re looking for solutions in real numbers or complex numbers, that’s usually a lot easier than looking for solutions that are natural or rational numbers.” The Beal conjecture requires solutions in natural numbers and further is of high degree. It is thus not surprising for the problem to be very challenging.

The Beal conjecture is one of several conjectures that closely resembled FLT. We will share some of them in later issues.

Formula to solve a cubic equation

Everyone knows the formula that gives the solution of a quadratic equation. This is the statement that for $a \neq 0$, the equation $ax^2 + bx + c = 0$ has the following two roots:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The ‘ $\pm$ ’ sign reflects the fact that the equation has two roots.

This familiar formula can be expressed in a simpler form. Since $a \neq 0$, we can divide the whole equation by a , following which we need symbols for only two coefficients. That is, we can write the equation in the following

form,

$$x^2 + bx + c = 0,$$

in which case the formula for the roots is this:

$$x = \frac{-b \pm \sqrt{b^2 - 4c}}{2}.$$

You may wonder, is there such a formula for the roots of a cubic equation? Indeed there is — but it is terribly complicated!

Just to illustrate the complexity of the formula, we display below the roots of the equation

$$x^3 - x - 1 = 0.$$

As it has degree 3, it has three roots. Here they are:

$$\begin{aligned} & \frac{1}{3} \sqrt[3]{\frac{27}{2} - \frac{3\sqrt{69}}{2}} + \frac{\sqrt[3]{\frac{1}{2}(9 + \sqrt{69})}}{3^{2/3}}, \\ & -\frac{1}{6} (1 - i\sqrt{3}) \sqrt[3]{\frac{27}{2} - \frac{3\sqrt{69}}{2}} - \frac{(1 + i\sqrt{3}) \sqrt[3]{\frac{1}{2}(9 + \sqrt{69})}}{2 \cdot 3^{2/3}}, \end{aligned}$$

and

$$-\frac{1}{6} (1 + i\sqrt{3}) \sqrt[3]{\frac{27}{2} - \frac{3\sqrt{69}}{2}} - \frac{(1 - i\sqrt{3}) \sqrt[3]{\frac{1}{2}(9 + \sqrt{69})}}{2 \cdot 3^{2/3}},$$

where $i = \sqrt{-1}$ is the imaginary unit. Whew! That was perhaps much more complicated than we expected. Time to take a pause now!

For more on the story of how cubic equations were cracked, you could refer to [6]. It is a fascinating story.

Further reading

- [1] Dana Pellegrino. “Pierre de Fermat” from
<https://sites.math.rutgers.edu/~cherlin/History/Papers2000/pellegrino.html>
- [2] Wikipedia. “Fermat’s Last Theorem” from https://en.wikipedia.org/wiki/Fermat's_Last_Theorem
- [3] O’Connor, J J; Robertson, E F. “Fermat’s last theorem” from
https://mathshistory.st-andrews.ac.uk/HistTopics/Fermat's_last_theorem/
- [4] PBS NOVA. “The Proof”. (The film discusses Andrew Wiles’s effort to prove Fermat’s Last Theorem.)
<https://www.pbs.org/wgbh/nova/proof/>
- [5] Wikipedia. “Beal conjecture” from
https://en.wikipedia.org/wiki/Beal_conjecture
- [6] Wikipedia. “Cubic equation” from
https://en.wikipedia.org/wiki/Cubic_equation

Pi on the Moon!

Adapted from: "Math Enthusiasts Unite to Have Rover Calculate Pi on the Moon." <https://www.scientificamerican.com/article/math-enthusiasts-unite-to-have-rover-calculate-pi-on-the-moon/>. Author: Emma R. Hasson (edited by Sarah Lewin Frasier)

Astrobotic's CubeRover, a shoebox-sized lunar rover set to launch later this year, will use its spare computing power to calculate the value of π in a novel way. This idea originated when Astrobotic engineers contacted Matt Parker, a math YouTuber and comedian known for his creative π calculations, to propose a meaningful use for the rover's unused computational resources. Parker devised a plan to calculate π on the Moon using random numbers generated from the rover's lunar data. These numbers could be applied to mathematical methods such as estimating π via random points inside a square or sphere, or through formulas involving the surface area of a sphere, paralleling the rover's movements on the Moon.



Figure 19: Astrobot's CubeRover is slated to land on the moon in late 2025. NASA/Kim Shiflett. Source: *Scientific American*, June 2025, "Math Enthusiasts Unite to Have Rover Calculate Pi on the Moon." <https://www.scientificamerican.com/article/math-enthusiasts-unite-to-have-rover-calculate-pi-on-the-moon/>

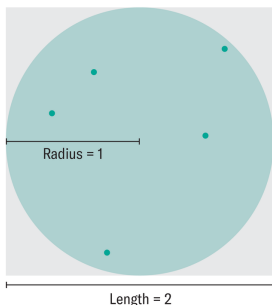
Though the project posed significant engineering challenges and required \$150,000 to integrate the code into the CubeRover without interfering with its primary mission, Parker turned to Kickstarter. Within four hours, his crowdfunding goal was surpassed, thanks to an enthusiastic online community of math lovers. Parker attributes this overwhelming support to the sense of community fostered by the Internet, which connects math enthusiasts who may have once felt isolated.

The project is not expected to yield groundbreaking scientific insights, but its purpose is to showcase math as an enjoyable and inclusive pursuit rather than something solely utilitarian. Parker emphasizes that math

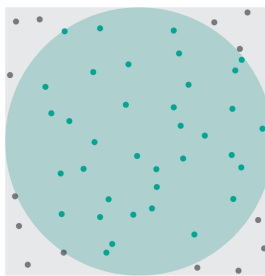
can be done for fun and that its cognitive and social benefits are best realized when approached as a creative activity. To further this message, he plans to involve students in the project, offering schools the opportunity to participate if their students calculate π by hand. Already, nearly 100 schools and over 2,000 students have joined the initiative.

When a circle with a radius of 1 is placed within a 2×2 square, randomly scattered points can be used to estimate the area of the circle.

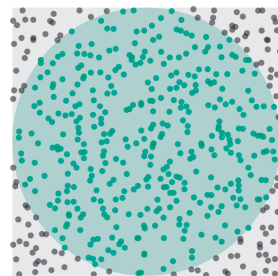
5 POINTS



50 POINTS



500 POINTS



Given the known area of the square (4), we can use the share of points that fall within the circle to infer the area of the circle and then use that value to calculate pi.

$$\frac{5 \text{ points in the circle}}{5 \text{ points total}} = 1$$

$$\text{Estimated value of } \pi = 1 \times 4 = 4$$

$$\frac{36 \text{ points in the circle}}{50 \text{ points total}} = 0.72$$

$$\text{Estimated value of } \pi = 0.72 \times 4 = 2.88$$

$$\frac{382 \text{ points in the circle}}{500 \text{ points total}} = 0.764$$

$$\text{Estimated value of } \pi = 0.764 \times 4 = 3.056$$

As more points are added, the solution gets closer to the actual value of pi (π).

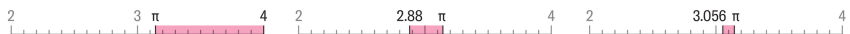


Figure 20: Credit: Amanda Montañez. *Scientific American*, June 2025, "Math Enthusiasts Unite to Have Rover Calculate Pi on the Moon." <https://www.scientificamerican.com/article/math-enthusiasts-unite-to-have-rover-calculate-pi-on-the-moon/>

Andrea Davis, project manager at Astrobotic, is excited about the educational aspect of the project, hoping it inspires students to see themselves as future engineers or space exploration contributors. She envisions the experience as a way for young people to connect with real-world lunar projects and recognize their potential to achieve similar feats in the future.

Rubin Observatory Data Flood Will Let the Universe Alert Astronomers 10 Million Times a Night

Adapted from: "Astronomers Brace for 10 Million Alerts a Night from Rubin Observatory" at <https://www.scientificamerican.com/article/rubin-observatory-data-flood-will-let-the-universe-alert-astronomers-10/?>. Author: Meghan Bartels (edited by Lee Billings)

The Vera C. Rubin Observatory in Chile has unveiled its first images, showcasing its vast field of view and exceptional depth. The observatory's groundbreaking capabilities lie in its ability to detect and alert astronomers to any changes in the night sky, producing an unprecedented volume

of data. It is expected to generate roughly 10,000 alerts per image and 10 million alerts per night, far exceeding manual data processing capacities. To manage this, the observatory will construct a static sky template to highlight changes, with automated algorithms correcting images and identifying differences. Scientists and amateurs can subscribe to “alert brokers” to receive tailored notifications about specific phenomena, such as supernovae or asteroids.

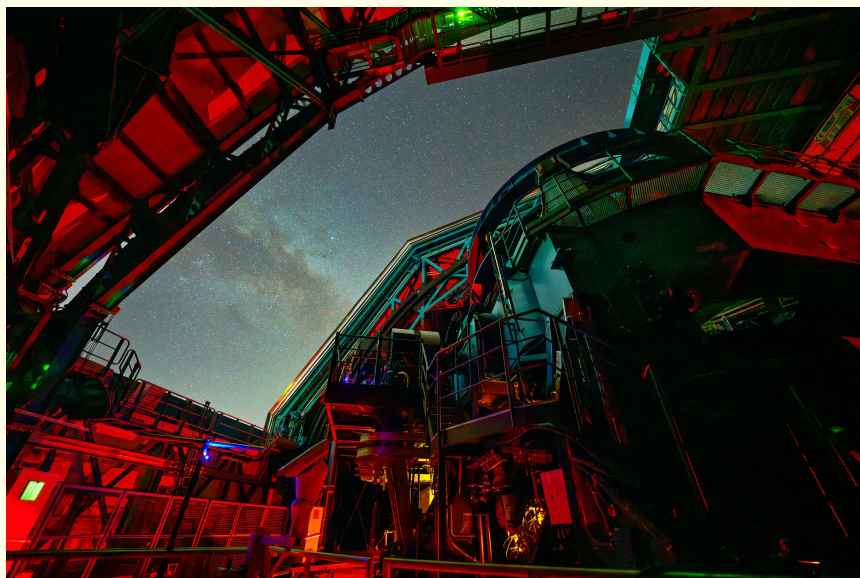


Figure 21: The telescope inside the dome of the NSF-DOE Vera C. Rubin Observatory. NSF-DOE Vera C. Rubin Observatory/H. Stockebrand (CC BY 4.0). Source: <https://www.scientificamerican.com/article/rubin-observatory-data-flood-will-let-the-universe-alert-astronomers-10/>

Over the next decade, the Rubin Observatory will conduct the Legacy Survey of Space and Time (LSST), capturing around 1,000 images nightly and scanning the entire visible Southern Hemisphere sky every 3-4 nights, creating the most detailed “movie of the heavens” ever conceived. Alongside, the telescope will remain adaptable to emerging scientific interests and unexpected events, such as gravitational-wave counterparts.

Rubin’s capabilities are particularly transformative in planetary science. It is expected to triple the known count of asteroids. For example, during the first two years of the survey, it could discover 70% of its predicted new objects. These discoveries, automatically reported to the Minor Planet Center, will allow astronomers to observe space rocks in real time.

Beyond rapid discoveries, the Rubin Observatory will advance the study of dark matter, the enigmatic substance that holds galaxies together. By measuring how dark matter distorts light from distant objects, astronomers can map its distribution across the universe. This capability fulfils a vision long pursued by Anthony Tyson, the observatory’s founding director, who had earlier proposed a similar project but was denied access to existing telescopes. His persistence led to the development of the Rubin Observatory, equipped with the largest digital camera in the world.

In addition to dark matter, the observatory will study phenomena like active galactic nuclei, detecting tens of millions of these energetic regions around supermassive black holes. Overall, Rubin is poised to revolutionize our understanding of the universe, providing rapid discoveries and profound insights into the cosmos. Scientists believe the LSST survey will not only answer existing questions but also uncover entirely new phenomena, fundamentally transforming our perception of the Universe.

Mathematicians are chasing a number that may reveal the edge of maths

Adapted from: “Mathematicians are chasing a number that may reveal the edge of maths” <https://www.newscientist.com/article/2487058-mathematicians-are-chasing-a-number-that-may-reveal-the-edge-of-maths/>. Author: Karmela Padavic-Callaghan

The Busy Beaver problem, a concept rooted in Alan Turing’s work on computability, examines the longest possible run-time of Turing machines with a given number of states. These maximum run-times, known as Busy Beaver numbers ($BB(n)$), grow at an extraordinary rate. For instance, $BB(1) = 1$, $BB(2) = 6$, and

$$BB(5) = 47,176,870.$$

But the exact value of $BB(6)$ remains unknown. Recent discoveries by the Busy Beaver Challenge, an online community of mathematicians, have set a new lower bound for $BB(6)$ that is unimaginably large, requiring advanced mathematical notation to describe.

Turing machines, which perform computations by following instructions to manipulate an infinite tape of 0s and 1s, serve as a simplified model for algorithms. For any given number of states, there are only finitely many possible Turing machines, but determining how long each machine will run before halting is extremely challenging.

The *Busy Beaver function* tracks the longest runtime for machines with a specific number of states, and the effort to determine $BB(6)$ has captivated contributors worldwide. In 2024, the Busy Beaver Challenge determined $BB(5)$ after four decades of effort, and now, a contributor known as “mxdys” has determined a lower bound for the number. The bound is most conveniently described using *tetration*, a mathematical operation involving iterated exponentiation. (See the appendix, below, for some details about the operation of tetration.) This contributor has shown that $BB(6)$ must be at least “2 tetrated to the 2 tetrated to the 2 tetrated to the 9”. This number is so vast that it dwarfs even the total number of particles in the universe. The scale is incomprehensible.

Beyond their sheer size, Busy Beaver numbers hold deep significance in mathematics at the foundational level. Alan Turing’s findings revealed that certain computational problems exceed the limits of mathematical predictability. Busy Beaver numbers make these abstract ideas tangible, allowing researchers to quantify where unpredictability may arise. Thus, the study of Busy Beaver numbers provides a concrete framework for understanding the limits of mathematical knowledge.

There could be a link between $BB(6)$ and the Collatz conjecture (we have written about this conjecture elsewhere in this issue). If a Turing machine responsible for $BB(6)$ could reproduce steps of the Collatz conjecture and halt, it might provide a computational proof for one version of the conjecture.

The Busy Beaver Challenge, launched in 2022 by computer scientist Tristan Stérin, has drawn participants from diverse backgrounds, including non-academics. Contributor Ligocki describes the Busy Beaver problem as a “scale for pondering the frontier of mathematical knowledge.” The challenge of exploring such vast and abstract numbers, while daunting, provides insight into previously unintelligible areas of mathematics.

While predicting the exact value of $BB(6)$ is impossible, recent progress suggests that further discoveries are imminent. Research may yield more breakthroughs, though the exact trajectory of the Busy Beaver problem remains uncertain.

The quest for $BB(6)$ illustrates beautifully the intersection of computation, mathematics, and philosophy, offering a glimpse into the limits of human understanding. Its study provides a framework for exploring the unknown frontiers of logic and computation.

Appendix: The operation of tetration

Tetration is an operation based on iterated, or repeated, exponentiation. No standard notation has been devised to represent this operation. The notation we shall use is the double arrow notation ($\uparrow\uparrow$) of Donald Knuth.

The number $a \uparrow\uparrow n$ is recursively defined as follows: $a \uparrow\uparrow 0 = 1$, and for $n > 1$,

$$a \uparrow\uparrow n = a^{a \uparrow\uparrow (n-1)}.$$

So we have,

$$a \uparrow\uparrow 1 = a,$$

$$a \uparrow\uparrow 2 = a^a,$$

$$a \uparrow\uparrow 3 = a^{a^a},$$

$$a \uparrow\uparrow 4 = a^{a^{a^a}},$$

and so on; $a \uparrow\uparrow n$ is read as “the n -th tetration of a ”; n is the ‘height’ of $a \uparrow\uparrow n$, while a is the ‘base’. We must keep in mind that

$$a^{a^a} \text{ means } a^{(a^a)} \text{ and not } (a^a)^a.$$

Examples

$$2 \uparrow\uparrow 1 = 2,$$

$$2 \uparrow\uparrow 2 = 2^2 = 4,$$

$$2 \uparrow\uparrow 3 = 2^4 = 16,$$

$$2 \uparrow\uparrow 4 = 2^{16} = 65536.$$

The next number in the sequence is

$$2 \uparrow\uparrow 5 = 2^{65536},$$

which has more than 19,700 digits.

Similarly,

$$3 \uparrow\uparrow 1 = 3,$$

$$3 \uparrow\uparrow 2 = 3^3 = 27,$$

$$3 \uparrow\uparrow 3 = 3^{27} = 7625597484987,$$

and the next number in the sequence is

$$3 \uparrow\uparrow 4 = 3^{7625597484987},$$

a number with over three and a half trillion digits! There is no possibility of ever writing down this number explicitly. The numbers generated via this simple-looking notation are staggering.

The lower bound for $BB(6)$ expressed using repeated exponentiation is,

$$BB(6) \geq 2^{2^{2^9}} = 2^{2^{512}}.$$

Readers could try estimating the size of this number for themselves.

Record-Breaking Results Bring Fusion Power Closer to Reality

<https://www.scientificamerican.com/article/record-breaking-results-bring-fusion-power-closer-to-reality/>. Author: By Tom Metcalfe (edited by Lee Billings)

The article discusses recent advancements and ongoing challenges in the pursuit of controlled nuclear fusion, a potential source of abundant and clean energy.

Fusion aims to replicate the process powering stars, using hydrogen isotopes like deuterium and tritium. While progress has been slow for decades, recent breakthroughs suggest that practical fusion power might be closer than previously believed.

Here are some of the key developments that have been achieved.

Magnetic Confinement Fusion (Stellarators and Tokamaks)

- In Germany's Wendelstein 7-X reactor, a record was set by maintaining a plasma — a superheated state of hydrogen gas — for 43 seconds using external superconducting magnets. This stellarator design offers stability and aims to operate continuously for extended periods.
- The competing JET reactor in the UK, a *tokamak*, managed to contain plasma for 60 seconds before its retirement in 2023. Tokamaks induce electric currents within the plasma, making them simpler but less stable than stellarators. A rivalry exists between these designs, with researchers debating which is better suited for future fusion reactors.

Inertial Confinement Fusion

The U.S.-based National Ignition Facility (NIF) achieved a milestone in 2022 by using giant lasers to ignite a fusion reaction in a pellet of hydrogen isotopes, producing more energy than consumed by the reaction itself. However, this method is impractical for electricity generation, as the lasers consume enormous amounts of energy and require lengthy recharging times.

Private Sector Involvement

Companies like General Fusion (Canada) and Commonwealth Fusion Systems (U.S.) are advancing alternative fusion technologies, such as magnetized target fusion and compact tokamaks. These projects aim to deliver electricity to the grid by the 2030s, driven by significant private investment.

Technological Challenges

- **Engineering Hurdles:** Both magnetic confinement and inertial confinement approaches face significant technical difficulties. For example, magnetic confinement requires precise control over plasma to prevent damage to reactor walls, while inertial confinement must scale up energy efficiency for continuous operation.
- **Energy Output:** For fusion power plants to be viable, they must sustain reactions at scale. For example, inertial fusion would require igniting multiple fuel pellets per second.
- **Superconducting Magnets:** These magnets, cooled near absolute zero, are critical for stabilizing plasma in magnetic confinement reactors. They offer greater control and are a key technology for future advancements.

Scientific and Practical Implications

The race between stellarators and tokamaks remains unresolved. Some researchers speculate that a hybrid design could combine the stability of stellarators with the simplicity of tokamaks, but such a solution is still in the early stages of exploration.

Despite skepticism about inertial confinement's practicality, its demonstration of net energy gain was a major scientific breakthrough, proving that controlled fusion is possible.

Future Outlook

Experts estimate that practical fusion power plants could become a reality within 15–20 years, though significant experimentation and testing remain. The private sector’s increasing involvement, coupled with advancements in superconducting magnet technology, may accelerate progress. However, both magnetic and laser-based approaches to fusion still face engineering challenges that must be resolved before fusion can become a reliable energy source.

In summary, while controlled nuclear fusion remains a scientific and engineering challenge, recent breakthroughs and growing investment suggest that this long-elusive “holy grail” of energy production could be closer to realization than ever before.

(This is a summary of the original article, produced using GPT-4o.)

Mathematicians Hunting Prime Numbers Discover New Patterns for Finding Them

Adapted from <https://www.scientificamerican.com/article/mathematicians-hunting-prime-numbers-discover-infinite-new-pattern-for/>. Author: Rachel Crowell (edited by Clara Moskowitz)

Mathematicians have made a significant breakthrough in the study of **prime numbers** by using a concept known as **integer partitions**, connecting two seemingly unrelated mathematical areas. Prime numbers, which are divisible only by 1 and themselves, are fundamental yet enigmatic. While identifying small primes is straightforward, discerning large primes becomes increasingly challenging. For example, the largest known prime, $2^{136279841} - 1$, contains over 41 million digits. Mathematicians aim not

only to identify individual primes but also to uncover patterns in their distribution.

Ken Ono of the University of Virginia, along with William Craig and Jan-Willem van Ittersum, has introduced a novel method for detecting primes. Their approach moves beyond traditional methods of factoring numbers. It offers infinitely many new definitions of what it means for a number to be prime. This work was recognized as a runner-up for a prestigious physical science prize.

Central to their discovery is the concept of *integer partitions*, which dates back to 18th-century mathematician Leonhard Euler. Integer partitions involve expressing a number as the sum of other numbers. For example, the number 5 has seven partitions:

$$5, \quad 4 + 1, \quad 3 + 2, \quad 3 + 1 + 1, \quad 2 + 2 + 1,$$

$$2 + 1 + 1 + 1, \quad 1 + 1 + 1 + 1 + 1.$$

Though partitions may seem elementary, they are revealed to be powerful tools for detecting primes. The researchers demonstrated that prime numbers are solutions to infinitely many polynomial equations in partition functions. These equations, known as Diophantine equations, can have integer or rational solutions. In essence, integer partitions serve as a new, natural way to identify prime numbers.

One specific equation provided by the researchers involves well-studied partition functions. If the equation holds true for an integer $n \geq 2$, then n is prime. The researchers showed that there are infinitely many such equations, providing an infinite array of new ways to define primes. Ono describes this discovery as “mind-blowing,” as it fundamentally broadens the understanding of primes.

Mathematicians, including Kathrin Bringmann from the University of Cologne, view the findings as groundbreaking. Bringmann suggests the research could inspire further exploration into the algebraic and analytic properties of combinatorial functions like partition functions. For instance, mathematicians might investigate whether similar methods could be used to study composite numbers or other mathematical sequences. This work exemplifies the deep interconnectedness of mathematics and could stimulate innovation across various subfields.

Despite this progress, many long-standing questions about primes remain unresolved, including the twin prime conjecture and Goldbach's conjecture. The twin prime conjecture posits the existence of infinitely many pairs of primes separated by two (e.g., 5 and 7, or 11 and 13). Goldbach's conjecture asserts that every even number greater than 2 can be expressed as the sum of two primes. Although Ono's team's findings do not directly address these problems, they represent a significant step forward in understanding prime numbers and their properties.

George Andrews, a mathematician at Pennsylvania State University, praises Ono as a visionary, noting that his work continues to shed new light on classical problems in mathematics. The team's discovery highlights the richness of mathematical connections and the potential for further breakthroughs in the study of primes.

(This is a summary of the original article, produced using GPT-4o.)

William Dalrymple's new book traces ancient India's role in spreading ideas

Adapted from: <https://www.theweek.in/theweek/leisure/2024/12/14/the-golden-road-how-ancient-india-transformed-the-world-william-dalrymple.html>. Author: Meera Suresh

The article discusses how William Dalrymple's book *The Golden Road: How Ancient India Transformed the World* retraces the influence of ancient India on global history. The narrative begins with an anecdote from the late 8th century AD, when a delegation from Sindh gifted a manuscript, *The Great Sindhind*, to Caliph Abu Ja'far al-Mansur in Baghdad. This manuscript, authored by Indian mathematician Brahmagupta, introduced concepts like zero, trigonometry, and algebra to the Islamic world, eventually influenc-

ing European mathematics through Leonardo of Pisa (who is better known as Fibonacci). The so-called “Arabic numerals” that Fibonacci introduced to Europe were, in fact, Indian in origin.

Dalrymple’s book explores India’s broader contributions beyond mathematics, highlighting the ancient trade, cultural, and religious exchanges that shaped civilizations across Asia and the world. Ancient Indian merchants, missionaries, monks, and artists paved a “Golden Road” stretching from the Red Sea to the Pacific. Dalrymple felt compelled to retell this underappreciated story, which he considers a debt to his younger self, inspired by his early fascination with ancient civilizations.

The inspiration for *The Golden Road* came during Dalrymple’s visit to the Ajanta Caves, where he observed restored Buddhist murals, prompting him to delve deeper into India’s ancient history. His research journey took him to sites across India and Southeast Asia, regions profoundly shaped by Indian influence. Dalrymple reveals India’s extensive trade links, particularly with the Roman Empire, as evidenced by the Muziris port in Kerala. Roman fleets transported Indian goods like black pepper, ivory, and silk, while Roman historian Pliny the Elder noted the economic drain caused by this trade. Archaeological findings, including the Muziris Papyrus, highlight the scale of commerce, equating the wealth generated by these exchanges to modern economic giants.

Dalrymple contrasts India’s trade impact with the more famous Silk Road, arguing that the latter did not exist in ancient times as popularly imagined. He asserts that the term “Silk Road” was created in the 19th century and only gained traction in modern discourse.

The book also emphasizes India’s export of religions, particularly Buddhism and Hinduism, as its most significant contribution. Indian monks

carried Sanskrit, literature, and religious teachings across Asia. Dalrymple vividly portrays figures like the monk Xuanzang, who brought Buddhism to China, and Empress Wu Zetian, who manipulated Buddhist monks for political legitimacy, marking a peak for Indian Buddhist influence in China.

Dalrymple attributes the neglect of India's historical contributions to the legacy of colonialism. British colonial attitudes, exemplified by figures like Thomas Macaulay, dismissed India's achievements to justify imperial domination. While early British scholars admired Indian culture, later colonial narratives undermined India's intellectual heritage, failing to recognize figures like Brahmagupta alongside Western mathematicians such as Pythagoras and Archimedes.

Through his book, Dalrymple seeks to illuminate the transformative role of ancient India in shaping world history and restore its rightful place in global narratives.

(This is a summary of the original article, produced using GPT-4o.)

Crows can recognize geometric regularity

Adapted from: <https://phys.org/news/2025-04-crows-geometric-regularity.html>?. Author: Bob Yirka

A trio of animal physiologists at the University of Tübingen, in Germany, has found that at least one species of crow has the ability to recognize geometric regularity. In their study published, the researchers conducted several experiments that involved testing crows on their ability to recognize geometric shapes.

Recognizing regularity in geometric shapes means being able to pick out one shape that is different from others in a group — picking out a plastic star, for example, when it is placed among several plastic moons. Testing for the ability to recognize geometric regularity has been done with many animals, including chimps and bonobos. Until now, this ability has never been observed in any creature except for humans.

Because of that, the team started with skepticism when they began testing carrion crows, using computer screens. The birds were asked to peck the outlier in a group; if they chose correctly, they got a food treat. The team chose to test carrion crows because prior experiments have shown them to have exceptional intelligence and mathematical capabilities.

All the birds tested caught on right away. Each could pick out a star if it was placed within a group of rectangles, for example. To determine their level of skill, the researchers started showing them objects that were only slightly different: one box that was tilted slightly, for example, among others that were not. Their accuracy went down, but it was still much higher than chance.

Continued testing showed that the birds had some understanding of right angles, parallel lines, and even some degree of symmetry.

The research team suggests their experiments were quite convincing — the crows were very obviously able to recognize geometric regularity. They note that the experiments also showed that the birds could do so with no prior learning. The findings strongly suggest that the ability exists in other animals as well.

(This is a summary of the original article, produced using GPT-4o.)

Google DeepMind and OpenAI's Achievements in the IMO

Remarks from Chat GPT

Introduction

Recent advancements in artificial intelligence (AI) have demonstrated remarkable capabilities in solving complex mathematical problems traditionally tackled by human prodigies. Among these advancements, Google DeepMind has achieved a groundbreaking milestone by securing the equivalent of a “Gold Medal” in a simulation of the **International Mathematical Olympiad (IMO)**. Concurrently, OpenAI has also made strides in mathematical problem-solving, showcasing AI’s expanding potential to master human-level reasoning in abstract domains.

Google DeepMind's Achievement

Google DeepMind, known for its innovations in AI technologies like AlphaFold and AlphaGo, has successfully trained a model capable of solving IMO-level problems. The IMO is a globally prestigious competition where high school students solve challenging mathematical problems in areas such as geometry, number theory, algebra, and combinatorics.

In this simulated competition, DeepMind's model achieved performance comparable to that of top-tier human participants, effectively earning the equivalent of a **Gold Medal**. This feat involved training an advanced neural network on extensive mathematical datasets, leveraging reinforcement learning and novel algorithms designed to mimic human problem-solving heuristics.

Key aspects of this achievement include:

- *Problem Understanding*. The AI demonstrated the ability to interpret and conceptualize problems written in natural language, translating them into mathematical terms for computation.
- *Logical Reasoning*. Unlike traditional AI models optimized for pattern recognition, this system exhibited reasoning akin to human mathematicians, applying step-by-step logic to derive solutions.
- *Generalization*. The model successfully solved problems it had not encountered during training, indicating progress in AI's ability to generalize knowledge in abstract domains.

DeepMind's accomplishment underscores the potential of AI not just as a tool for computation but as a collaborator capable of tackling some of the most intricate intellectual challenges.

OpenAI's Parallel Success

While DeepMind's achievement garnered significant attention, OpenAI has also demonstrated success in the field of mathematical reasoning. OpenAI's work focuses on developing large language models (LLMs), such as GPT-4, which have been fine-tuned to solve mathematical problems at various levels of complexity. OpenAI's approach leverages the following:

- *Language Understanding.* The models interpret mathematical problems within broader linguistic contexts, enabling them to engage in both computation and explanation.
- *Step-by-Step Reasoning.* Through fine-tuning and reinforcement learning, OpenAI's models have improved their ability to articulate multi-step solutions, mirroring the logical processes employed by mathematicians.
- *Collaboration with Humans.* OpenAI envisions its models as assistants to human researchers, helping them explore complex problems in mathematics, physics, and other scientific fields.

Although OpenAI's models have not directly competed in a simulated IMO, they have demonstrated comparable reasoning abilities in solving advanced mathematical problems, achieving results that rival DeepMind's.

Implications of These Achievements

These advancements have profound implications for mathematics, AI research, and broader scientific fields:

- *Mathematics Research.* The ability to solve IMO-level problems sug-

gests that AI can assist mathematicians in exploring new theorems, verifying proofs, and uncovering novel insights in unexplored areas of mathematics.

- *Education.* AI-powered tools could revolutionize mathematics education by providing personalized tutoring, generating problem sets, and offering step-by-step explanations tailored to individual learners.
- *Scientific Collaboration.* These breakthroughs highlight AI's potential as a collaborative partner in scientific research, capable of accelerating progress in fields like physics, engineering, and biology.

Ethical and Practical Considerations

While these achievements are exciting, they raise important ethical and practical questions:

- *Over-reliance on AI.* As AI becomes more integrated into problem-solving, ensuring that human expertise and creativity remain central to the process is critical.
- *Bias in Training Data.* The datasets used to train mathematical models must be carefully curated to avoid introducing biases that could limit their applicability.
- *Transparency.* AI systems must be designed to explain their reasoning to ensure their solutions are understandable and verifiable by human experts.

Conclusion

The recent achievements of Google DeepMind and OpenAI in mathematical reasoning represent a significant leap forward in AI's capabilities. By mastering problems of such complexity, these systems are not only pushing the boundaries of artificial intelligence but also opening new doors in mathematics, education, and scientific discovery. As these technologies continue to evolve, they will undoubtedly play an increasingly integral role in shaping the future of human knowledge and innovation.

Further Remarks — from Perplexity AI

Google DeepMind's Major Breakthrough at IMO 2025

In July 2025, Google DeepMind achieved a historic milestone by officially attaining the equivalent of a gold medal at the International Mathematical Olympiad (IMO)—the world's most prestigious mathematics competition for high school students. This feat marks the first time an artificial intelligence (AI) system has demonstrated mathematical reasoning and problem-solving prowess on par with the world's top young mathematicians in live competition settings.

How the Achievement Was Measured

During this year's IMO, hosted in Australia's Sunshine Coast, DeepMind's advanced Gemini model, running a specialized "Deep Think" system, participated as an official competitor. The model faced the same rigorous

conditions as human contestants: two 4.5-hour sessions, no internet or external tool access, and natural language statements of all problems. The Gemini system solved five out of six competition problems, achieving a total of 35 out of 42 points—the cutoff for a gold medal at the event. Only 67 out of 630 human participants reached this gold medal threshold, highlighting the Olympiad’s extreme difficulty.

The judge—who also graded the human contestants—evaluated Gemini’s handwritten, natural-language explanations and solutions, confirming that the system had reached the official gold medal standard. Notably, this year’s contest was the first where AI models were judged directly on their ability to process and solve problems entirely in natural language, without intermediate translation into formal logic or code. This is a significant leap from previous years, where AI systems, such as AlphaProof or AlphaGeometry, needed human experts to reformat problems for them.

Advancement Over Past AI Performance

DeepMind’s 2024 entry was only able to achieve a silver medal using systems that required formal, human-translated representations of math problems. Achieving gold using only natural language—mirroring how human contestants think and write—underscores dramatic improvements in the reasoning, communication, and language capabilities of today’s leading AI models.

OpenAI's Parallel Achievement

Just days before DeepMind's announcement, OpenAI revealed that its own state-of-the-art reasoning model had also achieved a gold medal-level score—solving five of six IMO problems for an identical 35 points—albeit outside of the official competition. OpenAI recreated contest conditions in-house (matching time limits, no web access or tools) and had its results independently validated by three past IMO medalists, though not by the official Olympiad jury.

OpenAI's approach leaned on a new method involving extensive “test-time compute;” the model was allowed to allocate massive computing resources to carefully explore many solution avenues in parallel—an approach described as “computationally very expensive.” The model tackled problems using natural language throughout, without math-specific prompting or shortcuts. Researchers at both companies stressed that these advances suggest AI systems are now “reasoning,” not merely regurgitating learned patterns or prompt templates.

The Significance and Broader Implications

These developments signal a new era for AI in symbolic reasoning and problem-solving. Both Gemini (DeepMind) and OpenAI's system have now matched the world's best high-school-aged human mathematicians under rigorous contest conditions for the first time ever. Experts highlight that formal recognition by IMO organizers is a crucial validation of DeepMind's accomplishment. In contrast, OpenAI's gold-level score, while independently verified, remains unofficial—a point acknowledged by both

companies and reflected in measured industry reactions.

The ability to solve IMO-level problems in natural language, without tailored translation or constraints, dramatically broadens the scope for AI in mathematics, science, and education. Researchers believe that tools capable of this kind of reasoning could soon assist in elite research, automate advanced education, and drive exploration at the frontiers of mathematics, physics, and computer science.

A New Benchmark for Artificial Intelligence

The achievements of Google DeepMind and OpenAI at the 2025 IMO represent a landmark in the evolution of artificial intelligence: for the first time, large language models have reached the gold medal standard in one of the world's most demanding intellectual arenas. While researchers urge caution and recognize that Olympiad mathematics and open-ended research are distinct, the new capabilities point to an accelerating convergence of AI and human cognitive performance at elite levels.

AI's Achievements in the IMO

— a rejoinder

From: "Mathematicians Question AI Performance at International Math Olympiad." Author: Emily Riehl. <https://www.scientificamerican.com/article/mathematicians-question-ai-performance-at-international-math-olympiad/>.

Mathematicians Question AI Performance at IMO

A defining memory from my senior year of high school was a nine-hour math exam with just six questions. Six of the top scorers won slots on the U.S. team for the International Math Olympiad (IMO), the world's longest running math competition for high school students. I didn't make the cut, but became a tenured mathematics professor anyway.

This year's olympiad, held last month on Australia's Sunshine Coast, had an unusual sideshow. While 110 students from around the world

went to work on complex math problems using pen and paper, several AI companies quietly tested new models in development on a computerized approximation of the exam. Right after the closing ceremonies, OpenAI and later Google DeepMind announced that their models earned (unofficial) gold medals for solving five of the six problems. Researchers like Sébastien Bubeck of OpenAI celebrated these models' successes as a "moon landing moment" by industry.

But are they? Is AI going to replace professional mathematicians? I'm still waiting for the proof.

The hype around this year's AI results is easy to understand because the olympiad is hard. To wit, in my senior year of high school, I set aside calculus and linear algebra to focus on olympiad-style problems, which were more of a challenge. Plus the cutting-edge models still in development did so much better at the exam than the commercial models already out there. In a parallel contest administered by MathArena.ai, Gemini 2.5 pro, Grok 4, o3 high, o4-mini high and DeepSeek R1 all failed to produce a single completely correct solution. It shows that AI models are getting smarter, their reasoning capabilities improving rather dramatically.

Yet I'm still not worried.

The latest models just got a good grade on a single test—as did many of the students—and a head-to-head comparison isn't entirely fair. The models often employ a "best-of-n" strategy, generating multiple solutions and then grading themselves to select the strongest. This is akin to having several students work independently, then get together to pick the best solution and submit only that one. If the human contestants were allowed this option, their scores would likely improve too.

Other mathematicians are similarly cautioning against the hype. IMO gold medalist Terence Tao (currently a mathematician at the University of California, Los Angeles) noted on Mastodon that what AI can do depends on what the testing methodology is. IMO president Gregor Dolinar said that the organization “cannot validate the methods [used by the AI models], including the amount of compute used or whether there was any human involvement, or whether the results can be reproduced.”

Besides, IMO exam questions don’t compare to the kinds of questions professional mathematicians try to answer, where it can take nine years, rather than nine hours, to solve a problem at the frontier of mathematical research. As Kevin Buzzard, a mathematics professor at Imperial College London, said in an online forum, “When I arrived in Cambridge UK as an undergraduate clutching my IMO gold medal I was in no position to help any of the research mathematicians there.”

These days, mathematical research can take more than one lifespan to acquire the right expertise. Like many of my colleagues, I’ve been tempted to try “vibe proving”—having a math chat with an LLM as one would with a colleague, asking “Is it true that...” followed by a technical mathematical conjecture. The chatbot often then supplies a clearly articulated argument that, in my experience, tends to be correct when it comes to standard topics but subtly wrong at the cutting edge. For example, every model I’ve asked has made the same subtle mistake in assuming that the theory of idempotents behaves the same for weak infinite-dimensional categories as it does for ordinary ones, something that human experts (trust me on this) in my field know to be false.

I’ll never trust an LLM—which at its core is just predicting what text will come next in a string of words, based on what’s in its dataset—to provide

a mathematical proof that I can't verify myself.

The good news is, we do have an automated mechanism for determining whether proofs can be trusted. Relatively recent tools called “proof assistants” are software programs (they don't use AI) designed to check whether a logical argument proves the stated claim. They are increasingly attracting attention from mathematicians like Tao, Buzzard and myself who want more assurance that our own proofs are correct. And they offer the potential to help democratize mathematics and even improve AI safety.

Suppose I received a letter, in unfamiliar handwriting, from Erode, a city in Tamil Nadu, India, purporting to contain a mathematical proof. Maybe its ideas are brilliant, or maybe they're nonsensical. I'd have to spend hours carefully studying every line, making sure the argument flowed step-by-step, before I'd be able to determine whether the conclusions are true or false.

But if the mathematical text were written in an appropriate computer syntax instead of natural language, a proof assistant could check the logic for me. A human mathematician, such as I, would then only need to understand the meaning of the technical terms in the theorem statement. In the case of Srinivasa Ramanujan, a generational mathematical genius who did hail from Erode, an expert did take the time to carefully decipher his letter. In 1913 Ramanujan wrote to the British mathematician G. H. Hardy with his ideas. Luckily, Hardy recognized Ramanujan's brilliance and invited him to Cambridge to collaborate, launching the career of one of the all-time mathematical “greats.”

What's interesting is that some of the AI IMO contestants submitted their answers in the language of the Lean computer proof assistant so that the computer program could automatically check for errors in their reasoning.

A start-up called Harmonic posted formal proofs generated by their model for five of the six problems, and ByteDance achieved a silver-medal level performance by solving four of the six problems. But the questions had to be written to accommodate the models' language limitations, and they still needed days to figure it out.

Still, formal proofs are uniquely trustworthy. While so-called “reasoning” models are prompted to break problems down into pieces and explain their “thinking” step by step, the output is as likely to produce an argument that sounds logical but isn't, as to constitute a genuine proof. By contrast, a proof assistant will not accept a proof unless it is fully precise and fully rigorous, justifying every step in its chain-of-thought. In some circumstances, a hand-waving or approximate solution is good enough, but when mathematical accuracy matters, we should demand that AI-generated proofs are formally verifiable.

Not every application of generative AI is so black and white, where humans with the right expertise can determine whether the results are correct or incorrect. In life, there is a lot of uncertainty and it's easy to make mistakes. As I learned in high school, one of the best things about math is the fact that you can prove definitively that some ideas are wrong. So, I'm happy to have an AI try to solve my personal math problems, but only if the results are formally verifiable. And we aren't quite there, yet.

A basketball problem

Here's a basketball problem for you to tackle (contributed by Dr. Ashika Jayanthi).

Do try to avoid looking at the solution too soon!

Problem

I went to the basketball court last evening and tried my luck at scoring baskets; I was hoping to achieve as high a score as possible. I took several shots at the hoop. My first shot was a failure but I persisted. Ultimately, I achieved a hit rate higher than 75%. Show that at some point, my hit rate must have been exactly 75%.

Solution

Let us first fix the notation. Just after the n -th shot let H_n be the number of successful hits ('baskets'), and let M_n be the number of misses. Then $H_1 = 0$ and $M_1 = 1$ (since the first shot was a failure).

Suppose the stated situation never occurs, i.e., at no stage is the hit rate exactly equal to 75%. This means that at no stage do we have $H_n = 3M_n$. We shall show that this supposition leads to a contradiction.

Consider the *first moment* when I achieve a hit rate strictly greater than 75%. Let this happen on shot n ; then $H_n > 3M_n$.

By assumption, the hit rate just after the previous shot is strictly less than 75%, i.e., $H_{n-1} < 3M_{n-1}$.

Since I must have scored a basket on shot n , it must be that

$$H_n = H_{n-1} + 1 \quad \text{and} \quad M_n = M_{n-1}.$$

Hence we have $H_n - 1 < 3M_n$.

Bringing together the different threads, we have the following:

$$H_n > 3M_n,$$

$$H_n < 3M_n + 1.$$

As H_n is an integer, and there cannot be an integer strictly more than $3M_n$ and also strictly less than $3M_n + 1$, the above inequalities cannot both be true. So we have reached a contradiction.

This shows that the assumption made at the start is untenable. So, the hit rate must have been exactly 75% at some stage. ■

Discussion

An important principle of problem-solving and the teaching-learning of mathematics is the following, formulated by Pólya:

Even fairly good students, when they have obtained the solution of the problem and written down neatly the argument, shut their books and look for something else. Doing so, they miss an important and instructive phase of the work. By looking back at the completed solution, by reconsidering and re-examining the result and the path that led to it, they could consolidate their knowledge and develop their ability to solve problems.

A good teacher should understand and impress on his students the view that no problem whatever is completely exhausted. There remains always something to do; with sufficient study and penetration, we could improve any solution, and, in any case, we can always improve our understanding of the solution.

Here we may ask: what is special about 75? Why 75? Why not some other figure? If we study the solution carefully we will see that its significance lies in the fact that $75/25 = 3$, an integer. Instead of 75, could we not have used 80, since $80/20 = 4$, an integer? Yes, of course.

Proceeding similarly, we obtain the following generalization.

Theorem

I go to the basketball court and try my luck at scoring baskets. I take several shots at the hoop. My first shot is a failure, but I persist. At any stage, let H be the number of successful hits and M the number of misses (so, after the first shot, $H = 0$ and $M = 1$). Let k be any given positive integer. Assume that at some stage I have sufficiently many successful hits so that $H > kM$. Then at some stage in the ‘game’ it must have been that $H = kM$.

Remark

This result may remind readers of the intermediate value theorem studied in a first course in calculus. The above result may be thought of as a discrete form of the intermediate value theorem!

Article contributed by the Editors

The Collatz problem

The Collatz problem, also known as the $3n + 1$ problem or the Syracuse problem, is an extremely famous unsolved conjecture in mathematics. Its formulation is simple but it has resisted solution for over 80 years despite attention from large numbers of mathematicians across the world.

The Problem

Start with any positive integer n . Apply the following iterative rule repeatedly:

- If n is even, divide it by 2.
- If n is odd, multiply it by 3 and add 1.
- Repeat for the answer thus obtained and continue in this manner.

The Collatz conjecture states that this process will always eventually reach 1, regardless of the starting number.

We may state the conjecture in terms of a sequence $a_1, a_2, a_3, \dots$ of positive integers as follows.

- Let $a_1 =$ any arbitrary positive integer.
- Compute successive members ($a_2, a_3, \dots$) of the sequence using the following rule:

$$a_{n+1} = \begin{cases} a_n/2, & \text{if } a_n \text{ is even,} \\ 3a_n + 1, & \text{if } a_n \text{ is odd.} \end{cases}.$$

- Then $a_n = 1$ for some positive integer n .

Examples

Starting with 5 This is what we get:

5, 16, 8, 4, 2, 1, 4, 2, 1, 4, 2, 1, 4, 2, ...

Observe how we reach 1 after 5 iterations.

Following this we are permanently stuck in the cycle (1,4,2). It is called a 3-cycle because it has three elements in it.

Starting with 3 This is what we get:

3, 10, 5, 16, 8, 4, 2, 1, 4, 2, 1, 4, 2, 1, 4, 2, ...

Observe that we reach 1 after 7 iterations, following which we are stuck in the same 3-cycle, (1,4,2).

Starting with 13 This is what we get:

13, 40, 20, 10, 5, 16, 8, 4, 2, 1, 4, 2, 1, 4, 2, ...

We reach 1 after 9 iterations.

Starting with 11 This is what we get:

11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1, 4, 2, ...

We reach 1 after 14 iterations.

Starting with 7 This is what we get:

7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1, 4, 2, ...

We reach 1 after 16 iterations.

Observe how in each case, after reaching 1, we get permanently stuck in the 3-cycle (1,4,2).

Current status of the conjecture

The conjecture has been computationally verified for all starting values up to approximately 2.95×10^{20} (as of 2020). No counterexample has yet been found.

Despite the overwhelming evidence in its favour, this does not constitute a proof of the conjecture. The Collatz conjecture remains an open problem.

What makes the conjecture so difficult to tackle?

The sequences generated by the Collatz iteration behave very non-intuitively. Perhaps this is because the iteration mixes multiplication ($n \mapsto 3n + 1$) with division ($n \mapsto n/2$). This mix somehow resists straightforward analysis.

It is interesting to note that Collatz sequences can grow very large before eventually decreasing. Some starting values produce peaks millions of times larger than the initial number!

The great mathematician Paul Erdős is reported to have said of the Collatz conjecture, “Mathematics may not be ready for such problems.” Despite this discouraging pronouncement, much research has gone into studying the problem and connections have been drawn to fields such as number theory, dynamical systems, and computational theory.

What is the significance of this conjecture?

The conjecture is tantalizing and most enticing because of its non-technical formulation. Whether it has implications in other fields of mathematics is not clear. But research has shown how simple iterative rules can generate very complex behaviour (as in the case of the Mandelbrot set).

Further reading

- [1] Wikipedia, “Collatz conjecture” from
https://en.wikipedia.org/wiki/Collatz_conjecture
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<https://www.quantamagazine.org/why-mathematicians-still-cant-solve-the-collatz-conjecture-20200922/>
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About the author

This note has been prepared by the editors using the LLM Claude-Opus-4.

Collatz pulls rabbits out of hats!

Article contributed by B Sury

The well known Fibonacci numbers $F_0 = 0, F_1 = 1, F_2 = 1, \dots$ are defined by the initial two values ($F_0 = 0, F_1 = 1$) and the recurrence relation $F_{n+1} = F_n + F_{n-1}$. (Some authors leave out the initial 0 and start directly with $F_1 = 1, F_2 = 1$.) The first few Fibonacci numbers are

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \dots$$

These omnipresent numbers can be 'seen' in nature too; for example, in the petals of a sunflower.

Originally defined by Fibonacci as a model of how the rabbit population is generated, it is veritably a trick of nature (one might say, Nature's way of pulling rabbits out of hats!) that so many natural phenomena produce the Fibonacci numbers.

Here we talk about a very curious way in which they are produced. There

is a famous unsolved problem known as the Collatz $3x + 1$ problem. (The preceding article in this issue describes the problem in detail.) Start with any natural number n . If it is even, divide it by 2; if it is odd, change it to $3n + 1$. Keep repeating this until you hit 1. The Collatz problem is to prove (or disprove) that indeed every n leads to 1 in a finite number of steps. It looks like an easy problem, but it has defied the attempts by mathematicians and computer scientists to solve it.

Let us try a few numbers. Evidently, any power of 2, say 2^k , leads to 1 in exactly k steps:

$$2^k \mapsto 2^{k-1} \mapsto \cdots 2 \mapsto 1.$$

What if we start with 3? We have

$$3 \mapsto 10 \mapsto 5 \mapsto 16 \mapsto 8 \mapsto 4 \mapsto 2 \mapsto 1.$$

This has taken 7 steps, while 7 takes 16 steps:

$$\begin{aligned} 7 &\mapsto 22 \mapsto 11 \mapsto 34 \mapsto 17 \mapsto 52 \\ &\mapsto 26 \mapsto 13 \mapsto 40 \mapsto 20 \mapsto 10 \\ &\mapsto 5 \mapsto 16 \mapsto 8 \mapsto 4 \mapsto 2 \mapsto 1. \end{aligned}$$

Exploring a simpler mapping

Analogous to the Collatz problem, let us consider the mapping

$$T(n) = \begin{cases} n/2, & \text{if } n \text{ is even,} \\ n + 1, & \text{if } n \text{ is odd.} \end{cases}$$

Let us try a few numbers, like earlier:

$$3 \mapsto 4 \mapsto 2 \mapsto 1 \mapsto 2 \mapsto 1 \mapsto \dots,$$

$$7 \mapsto 8 \mapsto 4 \mapsto 2 \mapsto 1 \mapsto 2 \mapsto 1 \mapsto \dots,$$

$$11 \mapsto 12 \mapsto 6 \mapsto 3 \mapsto 4 \mapsto 2 \mapsto 1 \mapsto 2 \mapsto 1 \mapsto \dots.$$

In contrast to the Collatz problem, it is easy to show that for every n , a finite number of repetitions of the transformation T leads to 1.

Indeed, if n is even, then we have,

$$n \mapsto \frac{n}{2},$$

which is strictly less than n for all natural numbers n .

If n is odd, then two repetitions of the transformation produces

$$n \mapsto n + 1 \mapsto \frac{n + 1}{2},$$

which is strictly less than n for $n > 1$.

Hence, whatever number n we start with, repeating the transformation keeps reducing the numbers (provided the number exceeds 1), so we must inevitably reach the smallest possible natural number, which is 1.

A surprising connection with the Fibonacci numbers

Above, we described the similarity of the T -map to the Collatz map, as well as some important differences.

But now comes something both unexpected and interesting.

To set this up, we define the **orbit** $\mathcal{O}(k)$ of any natural number k to be the sequence

$$k, T(k), T(T(k)), T(T(T(k))), \dots$$

For example, here is the orbit of 10:

$$\mathcal{O}(10) = 10, 5, 6, 3, 4, 2, 1, 2, 1, \dots$$

Here is the orbit of 15:

$$\mathcal{O}(15) = 15, 16, 8, 4, 2, 1, 2, 1, 2, 1, \dots$$

And here is the orbit of 100:

$$\mathcal{O}(100) = 100, 50, 25, 26, 13, 14, 7, 8, 4, 2, 1, 2, 1, \dots$$

Observe how in each case we reach 1 and then we get trapped eternally in the cycle $1, 2, 1, 2, \dots$

Question: After n iterations, how many positive integers end up at 1?

Here, we should keep in mind that we are not asking for the number of integers which lead to 1 in *exactly* n steps; we only ask that in n steps, we do end up at 1. (We may have reached 1 earlier too.)

To begin with, let us see how many integers n lead to 1 after 2 steps. We see that $n = 1$ and $n = 4$ end up in 1 after 2 steps:

$$1 \mapsto 2 \mapsto 1,$$

$$4 \mapsto 2 \mapsto 1.$$

A moment's thought tells us that these are the only two possibilities.

The only integers n leading to 1 in 3 steps are $n = 2$, $n = 3$, and $n = 8$:

$$2 \mapsto 1 \mapsto 2 \mapsto 1,$$

$$3 \mapsto 4 \mapsto 2 \mapsto 1,$$

$$8 \mapsto 4 \mapsto 2 \mapsto 1.$$

Similarly, there are 5 numbers leading to 1 in 4 steps, namely, $n = 1$, $n = 4$, $n = 6$, $n = 7$, and $n = 16$:

$$1 \mapsto 2 \mapsto 1 \mapsto 2 \mapsto 1,$$

$$4 \mapsto 2 \mapsto 1 \mapsto 2 \mapsto 1,$$

$$6 \mapsto 3 \mapsto 4 \mapsto 2 \mapsto 1,$$

$$7 \mapsto 8 \mapsto 4 \mapsto 2 \mapsto 1,$$

$$16 \mapsto 8 \mapsto 4 \mapsto 2 \mapsto 1.$$

The integers which lead to 1 in 5 steps are the following 8 numbers: 2, 3, 5, 8, 12, 14, 15, 32. We leave the verification to the reader.

Well, now it is a happy guess that the number of k which lead to 1 in n steps equals the $(n + 1)$ -th Fibonacci number F_{n+1} . We have just seen this for $n = 1, 2, 3, 4, 5$.

If this is true, how do we prove it?

Proof

To set this up, it is useful to examine the data closely. In the array below we have displayed the initial portions of the orbits $\mathcal{O}(1)$, $\mathcal{O}(2)$, $\mathcal{O}(3)$, ..., $\mathcal{O}(20)$, with all the 1's highlighted.

	k	1	2	3	4	5	6	7	8	9	10	11	12	13
$\mathcal{O}(1)$	1	2	1	2	1	2	1	2	1	2	1	2	1	2
$\mathcal{O}(2)$	2	1	2	1	2	1	2	1	2	1	2	1	2	1
$\mathcal{O}(3)$	3	4	2	1	2	1	2	1	2	1	2	1	2	1
$\mathcal{O}(4)$	4	2	1	2	1	2	1	2	1	2	1	2	1	2
$\mathcal{O}(5)$	5	6	3	4	2	1	2	1	2	1	2	1	2	1
$\mathcal{O}(6)$	6	3	4	2	1	2	1	2	1	2	1	2	1	2
$\mathcal{O}(7)$	7	8	4	2	1	2	1	2	1	2	1	2	1	2
$\mathcal{O}(8)$	8	4	2	1	2	1	2	1	2	1	2	1	2	1
$\mathcal{O}(9)$	9	10	5	6	3	4	2	1	2	1	2	1	2	1
$\mathcal{O}(10)$	10	5	6	3	4	2	1	2	1	2	1	2	1	2
$\mathcal{O}(11)$	11	12	6	3	4	2	1	2	1	2	1	2	1	2
$\mathcal{O}(12)$	12	6	3	4	2	1	2	1	2	1	2	1	2	1
$\mathcal{O}(13)$	13	14	7	8	4	2	1	2	1	2	1	2	1	2
$\mathcal{O}(14)$	14	7	8	4	2	1	2	1	2	1	2	1	2	1
$\mathcal{O}(15)$	15	16	8	4	2	1	2	1	2	1	2	1	2	1
$\mathcal{O}(16)$	16	8	4	2	1	2	1	2	1	2	1	2	1	2
$\mathcal{O}(17)$	17	18	9	10	5	6	3	4	2	1	2	1	2	1
$\mathcal{O}(18)$	18	9	10	5	6	3	4	2	1	2	1	2	1	2
$\mathcal{O}(19)$	19	20	10	5	6	3	4	2	1	2	1	2	1	2
$\mathcal{O}(20)$	20	10	5	6	3	4	2	1	2	1	2	1	2	1

Observe the two 1's in the column corresponding to $k = 2$; this corresponds

to $F_3 = 2$. Similarly, observe the three 1's in the column corresponding to $k = 3$ ($F_4 = 3$), and the five 1's in the column corresponding to $k = 4$ ($F_5 = 5$). We must now prove this property in general. That is, *we must find a way of matching the 1's that occur in any particular column with the 1's that occur in the previous two columns*. We do this as follows.

Let us start with a positive integer n and write, for convenience,

$$T_2(n) = T(T(n)), \quad T_3(n) = T_2(T(n)) = T(T(T(n))), \quad \dots$$

We must show that the equation $T_k(n) = 1$ gives exactly F_{k+1} values for n .

Consider some n with $T_k(n) = 1$.

Now n is either odd, or of the form $2 \pmod{4}$, or a multiple of 4.

We consider these three cases separately.

- Suppose n is odd. Then $T(n) = n + 1$, so

$$T_k(n) = T_{k-1}(T(n)) = T_{k-1}(n + 1).$$

Since $T_k(n) = 1$, it follows that $T_{k-1}(n + 1) = 1$.

We have thus matched the 1 in row n , column k with the 1 in row $n + 1$, column $k - 1$.

- Suppose n is of the form $2 \pmod{4}$. Then $T(n) = n/2$, so

$$T_k(n) = T_{k-1}(T(n)) = T_{k-1}(n/2).$$

Since $T_k(n) = 1$, it follows that $T_{k-1}(n/2) = 1$.

We have thus matched the 1 in row n , column k with the 1 in row $n/2$, column $k - 1$.

- Suppose n is a multiple of 4. Then $T(n) = n/2$ and $T_2(n) = n/4$, so

$$1 = T_k(n) = T_{k-1}(T(n)) = T_{k-2}(T_2(n)) = T_{k-2}(n/4).$$

Since $T_k(n) = 1$, it follows that $T_{k-2}(n/4) = 1$.

We have thus matched the 1 in row n , column k with the 1 in row $n/4$, column $k - 2$.

In other words, we have shown that the number of solutions n of $T_k(n) = 1$ is the sum of the number of solutions a of $T_{k-1}(a) = 1$ and the number of solutions b of $T_{k-2}(b) = 1$.

Expressed otherwise, we have matched the 1's that occur in any particular column with the 1's that occur in the previous two columns.

Inductively, assuming that the number of solutions a of $T_{k-1}(a) = 1$ is given by F_k , and the number of solutions b of $T_{k-2}(b) = 1$ is given by F_{k-1} (respectively), the number of solutions n of $T_k(n) = 1$ equals

$$F_k + F_{k-1} = F_{k+1},$$

by the definition of the Fibonacci numbers.

The claim follows by induction. ■

Further reading

[1] Wikipedia, "Fibonacci sequence" from

https://en.wikipedia.org/wiki/Fibonacci_sequence

[2] Wikipedia, "Collatz conjecture" from

https://en.wikipedia.org/wiki/Collatz_conjecture

- [3] Patrick Honner, “The Simple Math Problem We Still Can’t Solve,” Quanta Magazine, from <https://www.quantamagazine.org/why-mathematicians-still-cant-solve-the-collatz-conjecture-20200922/>

About the author

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Heron's Formula for Area of a Triangle – 5

We continue the series of articles on Heron's formula for area of a triangle, and now offer a second 'pure geometry' proof of Brahmagupta's theorem (it avoids the use of trigonometry).

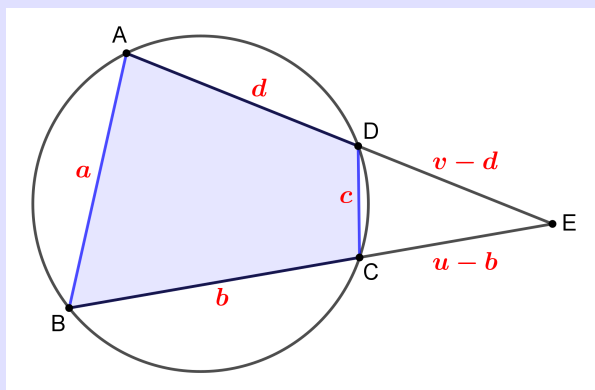
Statement of the theorem

Theorem (Brahmagupta). *Let $ABCD$ be a cyclic quadrilateral. Let its side-lengths be $AB = a$, $BC = b$, $CD = c$, and $DA = d$, and let $s = \frac{1}{2}(a + b + c + d)$ be the semi-perimeter. Then the area K of the quadrilateral is given by*

$$K = \sqrt{(s - a)(s - b)(s - c)(s - d)}. \quad (16)$$

Proof

The proof is based on the write-up in [1], which in turn is based on the proof given in the classic text [2]. It is a very straightforward proof; all it uses is Heron's formula for the area of a triangle, well-known properties of circles and similar triangles, and some algebra.



The lengths of various segments are as shown:

$$AB = a, BC = b$$

$$CD = c, DA = d$$

$$BE = u, AE = v$$

Figure 22: $ABCD$ is a cyclic quadrilateral. Opposite sides AD and BC meet at a point E which lies outside the circumscribing circle.

Heron's formula applied to $\triangle ABE$ yields:

$$4 \cdot [ABE] = \sqrt{(u + v + a)(u + v - a)(u - v + a)(v - u + a)}. \quad (17)$$

Drawing on the known properties of cyclic quadrilaterals, we know that $\triangle CDE$ is similar to $\triangle ABE$. Hence

$$\frac{CE}{AE} = \frac{DE}{BE} = \frac{CD}{AB}.$$

That is,

$$\frac{u-b}{v} = \frac{v-d}{u} = \frac{c}{a}. \quad (18)$$

From (18) we get

$$\frac{u-b}{c} = \frac{v}{a}, \quad \frac{v-d}{c} = \frac{u}{a}. \quad (19)$$

Addition of the two equalities in (19) yields

$$\frac{u+v}{c} - \frac{b+d}{c} = \frac{u+v}{a},$$

hence:

$$u+v = \frac{a(b+d)}{a-c}. \quad (20)$$

Subtraction of the two equalities in (19) yields

$$\frac{u-v}{c} - \frac{b-d}{c} = \frac{v-u}{a},$$

hence:

$$u-v = \frac{a(b-d)}{a+c}. \quad (21)$$

We now substitute these expressions into the expression for area of $\triangle ABE$ in (17). Since

$$\frac{a(b+d)}{a-c} + a = \frac{a(a+b-c+d)}{a-c},$$

$$\frac{a(b+d)}{a-c} - a = \frac{a(-a+b+c+d)}{a-c},$$

$$\frac{a(b-d)}{a+c} + a = \frac{a(a+b+c-d)}{a+c},$$

$$-\frac{a(b-d)}{a+c} + a = \frac{a(a-b+c+d)}{a+c},$$

we get:

$$\begin{aligned}
 & 4 \cdot [ABE] \\
 &= \frac{a^2}{a^2 - c^2} \sqrt{(a + b - c + d)(-a + b + c + d)(a + b + c - d)(a - b + c + d)}.
 \end{aligned} \tag{22}$$

Now invoke again the similarity of $\triangle ABE$ and $\triangle CDE$. This yields an area relation:

$$\frac{[CDE]}{[ABE]} = \frac{CD^2}{AB^2} = \frac{c^2}{a^2}. \tag{23}$$

Since the area of the quadrilateral is $K = [ABE] - [CDE]$, we get, from (23),

$$\frac{[ABCD]}{[ABE]} = \frac{a^2 - c^2}{a^2}. \tag{24}$$

Bringing together results (22) and (24) and writing $s = (a + b + c + d)/2$, we obtain the Brahmagupta formula:

$$[ABCD] = \sqrt{(s - a)(s - b)(s - c)(s - d)} \tag{25}$$

Though there is some algebra involved, the proof is straightforward and draws only on basic results. The reader will surely agree that it is a remarkable proof. ■

Area of a general convex quadrilateral

We conclude this article by quoting (without proof) a formula for the area of a general convex quadrilateral. It is due to the German mathematician Carl Anton Bretschneider, who discovered it in 1842; see [7].

Let $ABCD$ be a convex quadrilateral (not necessarily cyclic). Let its sides have lengths $AB = a$, $BC = b$, $CD = c$, $DA = d$. Let $s = (a + b + c + d)/2$ be the semi-perimeter of the quadrilateral. Let θ be half the sum of either pair of opposite angles, i.e., $\theta = (A + C)/2$ or $\theta = (B + D)/2$. (It does not matter which pair is used; you will see the reason for this in a moment.) Then the theorem states the following.

Theorem (Bretschneider).

$$[ABCD] = \sqrt{(s-a)(s-b)(s-c)(s-d) - abcd \cos^2 \theta}. \quad (26)$$

Remark

Since the angles $(A + C)/2$ and $(B + D)/2$ are supplementary, we have

$$\cos \frac{A + C}{2} = -\cos \frac{B + D}{2},$$

and as the cosine term occurs in squared form in the formula, it does not matter which pair of opposite angles is used. This explains the remark made above.

For a cyclic quadrilateral, both pairs of opposite angles add up to 180° , giving $\theta = 90^\circ$. Hence the cosine term vanishes and the formula reduces to the familiar Brahmagupta formula. So the Bretschneider formula is a true generalization of Brahmagupta's formula.

Further reading

[1] Alexander Bogomolny, "Brahmagupta's Formula and Theorem."

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- [2] R. A. Johnson, "Advanced Euclidean Geometry" (Dover, 2007).
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- [5] Wikipedia, "Ptolemy's theorem."
https://en.wikipedia.org/wiki/Ptolemy's_theorem
- [6] Jim Wilson (University of Georgia), "Ptolemy's theorem." <https://jwilson.coe.uga.edu/emt725/Ptolemy/Ptolemy.html>
- [7] Wikipedia, "Bretschneider's formula." https://en.wikipedia.org/wiki/Bretschneider%27s_formula

Circles in a Hotel Room

Article contributed by A.Ramachandran



Figure 23

Recently I came across this design in a hotel room (shown at the left). It set me thinking.

The first step was to have a verbal description of the design. It could go as follows.

Four circles are inscribed in a square, two large and two small. The two larger circles are of the same size; they touch each other and also the sides

of the square. Their centres lie on one of the diagonals. The two smaller circles too are of the same size. They do not touch each other, but touch both the larger circles and the sides of the square. Their centres lie on the other diagonal. The two diagonals are the only lines of symmetry of the figure.

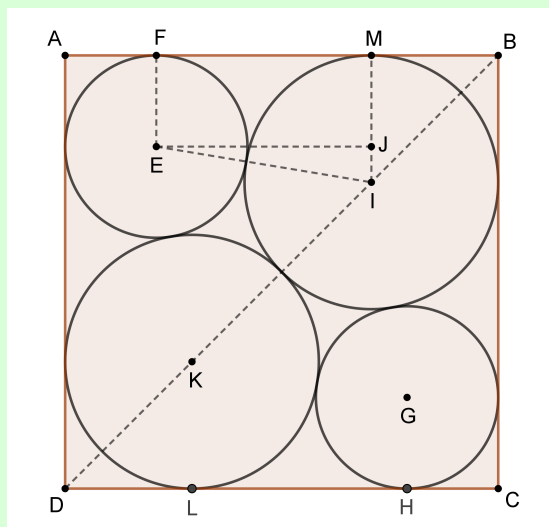


Figure 24

We ask the following questions, taking the side of the square to be 1:

- (a) What are the radii of the circles? (b) What are the coordinates of their centres (using some convenient system of coordinates)?

The reader should have a go at problems before reading on.

Imagine the square to be placed on the Cartesian plane, with the lower left corner at the origin and two sides on the axes.

We first obtain the value of R , the radius of the larger circle.

In the sketch, EF and IM are perpendicular to AB , so $IM = R$.

Also, $IB = R\sqrt{2} = KD$, and $IK = 2R$, so

$$BD = 2R + 2R\sqrt{2} = 2R(1 + \sqrt{2}) = \sqrt{2},$$

since the side of the square is 1. Hence $2R(1 + \sqrt{2}) = \sqrt{2}$, giving

$$R = \frac{\sqrt{2}}{2(1 + \sqrt{2})} = 1 - \frac{1}{\sqrt{2}} \approx 0.293 \quad (\text{to three decimal places}).$$

Let r be the radius of the smaller circles. We now obtain a relation between R and r . From the sketch we have,

$$EI = R + r, \quad JI = R - r, \quad EJ \perp MI,$$

so, $EJ^2 = EI^2 - JI^2 = 4Rr$, giving

$$EJ = 2\sqrt{Rr} = FM.$$

We have now,

$$AB = AF + FM + MB = r + 2\sqrt{Rr} + R = 1.$$

This gives rise to a very neat relationship:

$$\begin{aligned} (\sqrt{r} + \sqrt{R})^2 &= 1, \\ \therefore r &= (1 - \sqrt{R})^2. \end{aligned}$$

Since $R \approx 0.293$, we get $r \approx 0.210$.

It is now an easy task to obtain the coordinates of the centres of the circles. Starting from the circle at the top left and moving in a clockwise direction, the coordinates are, to three decimal places,

$$(0.210, 0.790), \quad (0.707, 0.707), \quad (0.790, 0.210), \quad (0.293, 0.293).$$

Another way to get the radius of the larger circle

We see that the circle with centre I , radius R is the incircle of $\triangle ABC$. Now we call upon the well-known relation

$$\text{Area of triangle} = \text{in-radius} \times \text{semi-perimeter}.$$

Using this we obtain the same value for R . (The area is $1/2$, and the semi-perimeter is $(2 + \sqrt{2})/2$; etc.)

One more task for the reader

A question that could be taken up now is how to construct such a figure using only a straightedge and compass.

Readers are welcome to respond.

About the author

A Ramachandran has had a long-standing interest in the teaching of mathematics and science. He studied physical science and mathematics at the undergraduate level and shifted to life science at the postgraduate level. He taught science, mathematics and geography to middle school students at Rishi Valley School for two decades. His other interests include the English language and Indian music. He may be contacted at archandran.53@gmail.com

An Algebraic Treat

Article contributed by Prithwijit De & Krishnendu Kundu

The genesis of this article lies in the following observation.

If a, b, c are positive real numbers such that

$$\frac{a}{b} = \frac{b}{c} = \frac{c}{a}$$

then $a = b = c$.

For, each fraction must be equal to

$$\frac{a + b + c}{a + b + c}$$

which equals 1, and the claim follows. (We have used the “addendo” principle here.)

Now we modify this a bit and ask the following question.

If a, b, c are positive real numbers such that

$$\frac{a^2}{b} = \frac{b^2}{c} = \frac{c^2}{a}.$$

Does it follow that $a = b = c$?

Here addendo does not work, for each fraction equals

$$\frac{a^2 + b^2 + c^2}{a + b + c}$$

which does not tell us much about the relationships between a , b and c .

Now, it is evident that if $a = b = c$ then

$$\frac{a^2}{b} = \frac{b^2}{c} = \frac{c^2}{a}.$$

If we have a firm belief that the converse is true, then we should be able to prove it in a satisfactory manner. One approach would be to assume the contrary and prove that it is false.

So, suppose $a \neq b$. Then either $a < b$ or $a > b$.

Case 1: $a < b$

If $a < b$ then $a^2 < b^2$ which implies

$$\frac{b}{c} = \frac{a^2}{b^2} < 1$$

and this in turn implies

$$\frac{c}{a} = \frac{b^2}{c^2} < 1.$$

Thus we obtain $a < b < c < a$, an impossibility.

Case 2: $a > b$

If $a > b$ then $a^2 > b^2$ which implies

$$\frac{b}{c} = \frac{a^2}{b^2} > 1$$

and this in turn implies

$$\frac{c}{a} = \frac{b^2}{c^2} > 1.$$

Thus we obtain $a > b > c > a$, again an impossibility. Hence $a = b$ which yields $b = c$ and hence $a = b = c$. ■

A direct solution is possible but not without taking recourse to some standard inequalities. Here is one possible way of proving $a = b = c$ using the A.M - G.M inequality which relies on the clever use of the inequality

$$3(a^2 + b^2 + c^2) \geq (a + b + c)^2.$$

We have

$$\frac{a^2}{b} = \frac{b^2}{c} = \frac{c^2}{a} = \frac{a^2 + b^2 + c^2}{a + b + c} \geq \frac{(a + b + c)^2}{3(a + b + c)} = \frac{a + b + c}{3}.$$

and

$$\frac{a^2}{b} = \frac{b^2}{c} = \frac{c^2}{a} = \left(\frac{a^2}{b} \cdot \frac{b^2}{c} \cdot \frac{c^2}{a} \right)^{1/3} = (abc)^{1/3}$$

whence

$$(abc)^{1/3} \geq \frac{a + b + c}{3} \geq (abc)^{1/3}$$

where the right hand inequality is the well-known AM-GM inequality. The upshot of the above string of inequalities is that we have equality in the AM - GM inequality and this forces

$$a = b = c.$$

Yet another way to solve this problem is to rewrite the given equalities as

$$a^3 = bc^2, \quad b^3 = ca^2, \quad c^3 = ab^2.$$

Then with a bit of algebra we arrive at

$$a^3 = b(b^3/a^2)^2$$

which is same as

$$a^7 = b^7.$$

Similarly, $b^7 = c^7$. So,

$$a^7 = b^7 = c^7$$

whence $a = b$ because

$$a^7 - b^7 = (a - b)(a^6 + a^5b + \cdots + ab^5 + b^6) = 0$$

implies $a = b$ as the other factor is positive.

Similarly $b = c$, and hence $a = b = c$. ■

The reader may have noticed that the problem can be extended to $n(> 3)$ variables $a_1, a_2, \dots, a_n$. The statement would read:

If the positive real numbers $a_1, a_2, \dots, a_n$ are such that

$$\frac{a_1^2}{a_2} = \frac{a_2^2}{a_3} = \cdots = \frac{a_{n-1}^2}{a_n} = \frac{a_n^2}{a_1}$$

then $a_1 = a_2 = \cdots = a_n$.

In the same vein we may consider the following problem.

If the sides a, b, c of a triangle are such that

$$\frac{a^2}{b^2 - bc + c^2} = \frac{b^2}{c^2 - ca + a^2} = \frac{c^2}{a^2 - ab + b^2}$$

then does it follow that $a = b = c$?

If in a triangle with sides a, b, c , the angle opposite the side a is 60° then the cosine rule yields

$$a^2 = b^2 + c^2 - 2bc \cos 60^\circ = b^2 - bc + c^2.$$

Thus, if

$$\frac{a^2}{b^2 - bc + c^2} = \frac{b^2}{c^2 - ca + a^2} = \frac{c^2}{a^2 - ab + b^2} = 1,$$

then we immediately see that $a = b = c$ because each angle of the triangle is 60° .

As earlier, we assume that $a \neq b$ and divide the analysis into two cases.

Case 1: $a < b$

In this case

$$\frac{a^2}{b^2} = \frac{b^2 - bc + c^2}{c^2 - ca + a^2} = \frac{(b - c)^2 + bc}{(c - a)^2 + ca} < 1$$

whence

$$(b - c)^2 - (c - a)^2 + c(b - a) = (b - a)(a + b - c) < 0$$

which implies $b < a$ since by triangle inequality we know that $a + b - c > 0$. A contradiction.

Case 2: $a > b$

Proceeding as above we end up with

$$(b - a)(a + b - c) > 0$$

which implies $a < b$. A contradiction. Hence $a = b = c$. ■

Hence $a = b = x$ (say). Then

$$\frac{x^2}{c^2 - cx + x^2} = \frac{c^2}{x^2 - x^2 + x^2}$$

whence

$$(c - x)(x^3 + cx^2 + c^3) = 0$$

implying $c = x$. Thus $a = b = c$.

Alternatively, if we assume that a, b, c are pairwise distinct then from

$$\frac{a^2}{b^2 - bc + c^2} = \frac{b^2}{c^2 - ca + a^2}$$

we obtain

$$(a - b)\{(a + b)(a^2 + b^2 + c^2) - c(a^2 + ab + b^2)\} = 0$$

whence

$$\frac{a^2 + b^2 + c^2}{a^2 + ab + b^2} = \frac{c}{a + b} < 1$$

which implies $c^2 < ab$. Similarly we can obtain $a^2 < bc$ and $b^2 < ca$. Combining the three inequalities we get

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2} \left((a - b)^2 + (b - c)^2 + (c - a)^2 \right) < 0,$$

a contradiction.

We close with two problems for the reader.

1. If a, b, c are positive real numbers such that

$$\frac{a^2}{b^2 + bc + c^2} = \frac{b^2}{c^2 + ca + a^2} = \frac{c^2}{a^2 + ab + b^2},$$

can a, b, c be pairwise distinct?

2. If a, b, c are positive real numbers such that

$$\frac{a^2}{b(c + a)} = \frac{b^2}{c(a + b)} = \frac{c^2}{a(b + c)},$$

does it follow that $a = b = c$?

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Call for Articles

We invite mathematics teachers and practitioners to contribute short articles to The Compass.

Articles could deal with mathematical content or the teaching of mathematics (classes 6 to 12). They could feature real-life classroom experiences with projects and explorations.

They could be about the use of audiovisual material or social media in the mathematics classroom.

They could feature reflections on the teaching of mathematics, or on the nature of mathematics.

Articles about the use of software are particularly welcome, as are articles dealing with non-routine problems and classroom experiences with problem-solving.

In short, anything at all to do with school mathematics!